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Quality over quantity: a data-centric survey of annotation errors in object detection datasets

Adnan Hussain[1] · Kaleem Ullah[1] · Muhammad Afaq[1] · Muhammad Munsif[1] · Altaf Hussain[1] · Sung Wook Baik[1]

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Abstract

In recent years, object detection has become a cornerstone of many computer vision applications, relying heavily on the availability of high-quality annotated datasets. However, even widely used benchmarks often suffer from annotation issues such as inaccurate bounding boxes, misclassified objects, and missing labels. These annotation errors, especially localization errors, can greatly affect the training and evaluation of detection models. In this survey, we provide a data-centric and comprehensive review of existing methods for identifying and analyzing errors in object detection datasets. We examine the main components of error detection workflows, including annotation error taxonomies and model-agnostic detection techniques. In addition, we develop a standardized categorization of annotation error types specific to object detection, providing a foundation for consistent analysis and comparison across studies. We also perform manual inspections of selected benchmark datasets to observe and quantify common annotation errors in practice. Moreover, the survey highlights the datasets used for evaluating error detection methods and compares their scope and inherent challenges. Finally, we summarize the types of annotation errors found in existing benchmarks and provide recommendations for future research to enhance dataset quality and reliability in object detection.

Keywords Object detection · Computer vision · Localization errors · Error detection · Dataset evaluation

1 Introduction

Object detection (OD) is a fundamental task in computer vision that identifies and locates objects in images. It has a wide range of applications, including surveillance systems (Nimma et al. 2025; Namana and Kumar 2025; Wu 2025; Abba et al. 2024; Jain et al. 2024) medical imaging (He et al. 2025; Sobek et al. 2024; John et al. 2025; Xu et al. 2025; Patro et al. 2025; Rezaei et al. 2025), autonomous driving (Zhao et al. 2025; Saeedizadeh et al. 2025; Yu 2025; Tahir et al. 2025; Zhang et al. 2025; Anguchamy and Palanisamy 2025; Cheng et al. 2025), and robotics (Liu et al. 2023; Mukherjee et al. 2025; Dai et al. 2024; Hong et al. 2024; Wu

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et al. 2024; Lim et al. 2024; Manakitsa et al. 2024). Recent advances in deep learning have produced highly accurate and efficient detectors (Suthaharan 2016; Schapire 2013; Girshick et al. 2014) yet their performance and reliability fundamentally depend on the quality of the underlying datasets, not only on model design. Widely used OD benchmarks such as MS COCO, open images, Pascal VOC, LVIS, Object365, FSOD, and DOTA have enabled rapid progress, but they also contain systematic annotation errors. These include missing or incomplete object labels, inaccurate or misaligned bounding boxes, inconsistent treatment of occlusion and truncation, ambiguous category definitions, class imbalance, and distributional bias. Such errors introduce label noise during training, corrupt evaluation metrics, and can lead to misleading conclusions about model performance, especially for long-tailed categories, few-shot settings, or safety-critical applications.

In practice, annotation errors and prediction errors remain two intertwined yet underexplored sources of failure in OD systems. Annotation errors arise from human mistakes, ambiguous labeling guidelines, limitations of annotation tools, or insufficient quality control during dataset construction. Prediction errors stem from the model's own failures, but are also entangled with noisy supervision, making it difficult to attribute errors purely to the model. Addressing these issues requires data-centric methodologies that can detect, validate, and correct annotation noise at scale, rather than focusing solely on new architectures or loss functions.

In response to these issues, there is an increasing number of works describing methods for detecting, validating, and correcting annotation errors in OD datasets, using methods that range from full manual inspection to weakly supervised, semi-supervised, and fully automatic pipelines. Manual approaches require an expert review of a sample of labels to identify unclear or problematic labels, and unverified labeling or mistakes are cross-checked. Weakly-supervised (Wang et al. 2022; Diba et al. 2017; Bilen and Vedaldi 2016), and semi-supervised (Zhu et al. 2025; Kim et al. 2025; Liu et al. 2024; Zhong et al. 2024; Han et al. 2024) include limited human supervision of the annotation, using a large pool of unlabeled images to either refine the labels or find images and/or instances that were not previously annotated. Fully automatic implementation (Liao et al. 2024; Tkachenko et al. 2023; Wu et al. 2023; Liu et al. 2022; Chachula et al. 2022) does not rely on human supervision, including methods that use model predictions (potential errors), loss patterns (poor predictions could be annotator problems), or self-training signals (through iterations of being trained with the corrected annotation). Ultimately, all these approaches aim to improve the quality of the dataset, and consequently, the OD model's robustness and trustworthiness.

Despite extensive progress in OD, most existing surveys remain model-centric. They primarily address detector architectures, training strategies, performance metrics, and task-level challenges such as class imbalance, small object detection, domain adaptation, open-world detection, and adversarial robustness. To provide readers with an understanding of the most recent advancements in the field of OD research, we have summarized the collection of survey articles from the previous five years (2021 to 2025), presented in Table 1. Where Oksuz et al. (2020) review the imbalance issues and provide a taxonomy of eight different challenges under four main categories, i.e., class imbalance, scale imbalance, spatial imbalance, and object imbalance. But does not address or analyze label or prediction errors in datasets or models. Antonelli et al. (2022) provide a detailed review on Few-Short object detection challenges, methods including data augmentation, metric learning, meta-learning, and transfer learning, and also cover datasets and evaluation metrics. But

Table 1 A summary and comparison of the existing surveys in the object detection field

#	Title	Year	Venue	Main contribution
1	Oksuz et al. (2020)	2021	TPAMI	This survey tackles the imbalance issue in object detection and presents a taxonomy of eight challenges across class, scale, spatial, and object categories, evaluating solutions from architecture, data, and loss design.
2	Antonelli et al. (2022)	2022	ACM (CSUR)	This survey covers few-shot OD, its challenges, and methods. It also reviews benchmarks, evaluation metrics, and datasets.
3	Huang et al. (2022)	2022	TPAMI	This paper reviews few-shot and self-supervised OD methods. It lacks a comprehensive summary of datasets and performance comparisons for SSOD and FSOD.
4	Li et al. (2023)	2023	TPAMI	This survey explores knowledge distillation-based object detection models, including concepts, tasks, and techniques like soft label, feature, self, and multi-teacher distillation.
5	Zou et al. (2023)	2023	Proceedings of the IEEE	This survey traces the evolution of OD from 1990 to 2022, from handcrafted methods to deep learning models like CNNs and Transformers.
6	Sun et al. (2024)	2024	EAAI	This survey covers the evolution of object detection from traditional methods to CNNs and transformers, including YOLO, SSD, Faster R-CNN, DETR, and key datasets and metrics.
7	Liang et al. (2024)	2024	TPAMI	This paper introduces a four-quadrant challenge framework and evaluates object detectors, discussing architectural limitations, solutions, and benchmarking on multiple datasets.
8	Oza et al. (2023)	2024	TPAMI	This paper reviews unsupervised domain adaptation methods for object detectors, covering strategies like adversarial learning, pseudo-labeling, image translation, and more, with taxonomy and future research insights.
9	Nguyen et al. (2025)	2025	TNNLS	This work analyzes adversarial attacks on object detection, providing a new taxonomy and experimental comparisons of detectors like DeTR and GLIP.
10	Li et al. (2024)	2025	TCSVT	This survey reviews Open World Object Detection (OWOD), focusing on detecting known, unknown, and novel objects with incremental learning, and comparing approaches across datasets.
11	Our	2026	Ours	Our survey paper aims to provide a comprehensive review of the taxonomy of errors in OD datasets. We examine the current methods used to identify different types of errors, assess how these methods validate their findings, and discuss their applicability to real-world tasks. Additionally, we critically evaluated the gaps in the literature, highlighting instances where some studies fail to verify the effectiveness of error detection, which limits their practical application. Finally, we suggested directions for future research.

does not address label or prediction error analysis or dataset annotation quality issues that impact detection performance. Similarly, Huang et al. (2022) analyze the few-shot learning and self-supervised methods, including definitions, benchmarks, major techniques for finetuning, and modulation-based approaches. But does not analyze or categorize works addressing label or prediction error sources or the impact of annotation quality on detection performance. Li et al. (2023) summarized the knowledge distillation-based models, their concepts, and techniques. Zou et al. (2023) review the evaluation of OD techniques from 1990 to 2022. But does not analyze or classify works in terms of label or prediction errors, nor address dataset annotation quality or error propagation in detection pipelines. Sun et al. (2024) explored the evaluation of different methods from conventional methods to advanced deep learning methods, datasets, and evaluation metrics. They divide detectors into two major stages, one-stage and two-stage detectors, and discuss the major developments of each stage of detectors. Liang et al. (2024) conducted a systematic review to provide a novel four-quadrant challenge framework (out-of-domain, out-of-category, robust learning, and incremental learning), which is introduced along with a thorough evaluation of object detectors in an open environment. It discusses architectural limitations of existing detectors, reviews state-of-the-art solutions, and benchmarks performance on multiple datasets. But does not address the challenges and solutions related to label or prediction errors, annotation noise, or error correction in detection pipelines. Oza et al. (2023) provided a comprehensive review of the recent methods and challenges of unsupervised adoption of Object detectors. They categorized the methods into six strategies: adversarial feature learning, pseudo-label self-training, image-to-image translation, domain randomization, mean teacher training, and graph reasoning. They do not address the issue of label or prediction errors arising from incorrect annotations or model misclassifications. Nguyen et al. (2025) analyzed and assessed adversarial attacks, providing a new taxonomy for identifying attack types and analyzing adaptability. It examines digital and physical attacks, white-, gray-, and black-box scenarios, attack components (e.g., classification score, NMS), and perturbation characteristics (e.g., patch versus universal attacks). They also provide a comparison on various conventional to modern detectors, such as the DeTR and the GLIP. Li et al. (2024) presented an overview of open world object detection (OWOD), highlighting detection of known, unknown, and new objects, providing incremental learning without catastrophic forgetting. Open Set Recognition and Incremental Learning are key framework areas discussed. It classifies approaches within OWOD (pseudo-labeling, class-agnostic, metric learning) and evaluates them on different datasets and metrics.

The existing reviews regarding the OD task (Oksuz et al. 2020; Antonelli et al. 2022; Huang et al. 2022; Li et al. 2023; Zou et al. 2023; Sun et al. 2024; Liang et al. 2024; Oza et al. 2023; Nguyen et al. 2025; Li et al. 2024), show the lack of thorough and systematic reviews of the approaches that discuss the methods used for annotation errors. The existing surveys mostly focus on the training methods, algorithms, evaluation metrics, and challenges related to object detection tasks. The identification and evaluation of annotation errors is a highly relevant topic of research, although it receives little attention compared to the creation of algorithms for object detectors and learning strategies. Manually identifying these errors is a laborious and time-consuming task that requires automated and scalable solutions. This encourages us to give a complete overview of all approaches proposed in recent years that are explicitly intended to address annotation errors. The major contributions of this survey are summarized as follows:

- We present the very first survey of the detection and validation of label and prediction errors (DVLPE) in object detection datasets. To the best of our knowledge, there is no existing survey in the DVLPE in OD literature, and it lacks the attention of researchers. With this motivation, we present a comprehensive and compact survey of all the existing methods related to the detection and validation of annotation errors.
- We present a focused survey on dataset evaluation methods, specifically targeting the quality and structure of datasets commonly used in the detection fields. To the best of our knowledge, the evaluation of datasets has not been given adequate attention in the literature, despite its critical impact on model performance. With this motivation, we

provide a comprehensive review of existing approaches for assessing dataset quality and relevance.

- In this survey, we develop a standardized categorization of annotation errors specific to object detection datasets through our detailed manual inspection of publicly available benchmarks. We identify and analyze common issues, including inaccurate bounding boxes, misclassified objects, missing annotations, class imbalance, and distributional bias. We further examine the implications of these errors on model performance and generalizability across object detection tasks.
- In this survey, we cover trends in DVLPE literature, the distribution based on publishers, citations, types of research papers, and application-wise scattering of these approaches, along with a taxonomy of the detection and validation of annotation errors methods. Further, this survey provides results of all the queries for searching related papers in various repositories and also offers remarks about the selection process of the retrieved papers in the survey.
- This survey explores the current methods of DVLPE, explores the datasets, challenges, and concludes with the overall literature. Finally, our survey provides recommendations and future research directions for further exploration of the DVLPE field.

The rest of the paper is organized as follows: the scope, outline, and coverage of the survey are discussed in Sects. 2, 3 investigates the existing methods, and the taxonomy of the DVLPE methods. The existing datasets used for the object detection task are provided in Sect. 4. The errors in such datasets highlight challenges of DVLPE methods, and provide future direction in Sect. 5. Finally, the survey is concluded in Sect. 6.

2 Scope and outline of the survey

This survey covers articles from various top conferences, workshops, and journals on DVLPE from diverse repositories including IEEE, Springer, Elsevier, CVPR, ICCV, etc., using different combinations of the keywords, i.e., "errors in object detection", "object detection", "object detection datasets", "errors in object detection datasets", "localization errors", "annotation errors", "label errors", etc. We have searched all papers published between 2016 and 2025. Overall, papers were explored through various searches conducted in multiple repositories. Distribution of DVLPE literature over a number of published articles per year, citations, papers published, and article types as shown in Fig. 1. The year-wise number of published articles is shown in Fig. 1a, and the number of citations per year is presented in Fig. 1b. A ratio of related papers published in different venues is represented in a pie chart, Fig. 1c. Conference papers, journals, and arxiv/workshop papers are visualized in Fig. 1d. This demonstrates the overall trend of the literature.

In the DVLPE literature, most papers are published in IEEE journals and top conferences. The year-wise trend in this domain is visualized in Fig. 2a, covering the overall literature from 2016 to 2025. In the starting year of $2016-17$, the literature focuses on label noise errors targeting the Pascal VOC, MS-COCO, and Open Image datasets, using semi-supervised, fully-supervised, and weakly supervised techniques. Similarly, in $2018-19$, researchers concentrated on other types of annotation errors, such as localization uncertainty, missing class labels, and overlapping bounding box errors, with a focus on additional

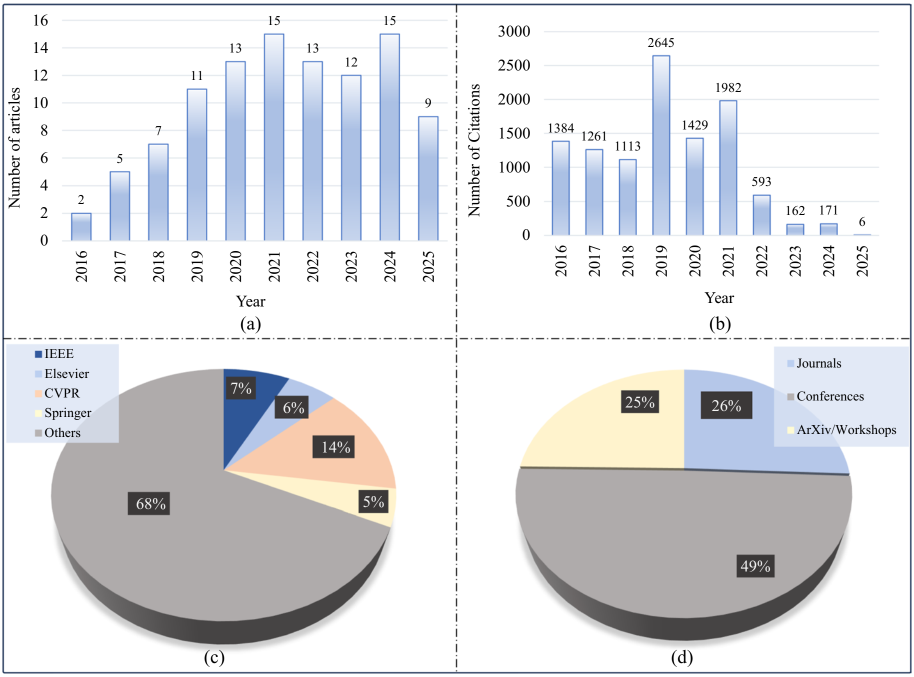

Fig. 1 Statistics of the covered literature: **a** year-wise DVLPE publications to date, **b** citation-wise distribution of the research papers, **c** publisher-wise distribution of the research papers, and **d** categorization of papers selected from journals, top conferences, and other sources

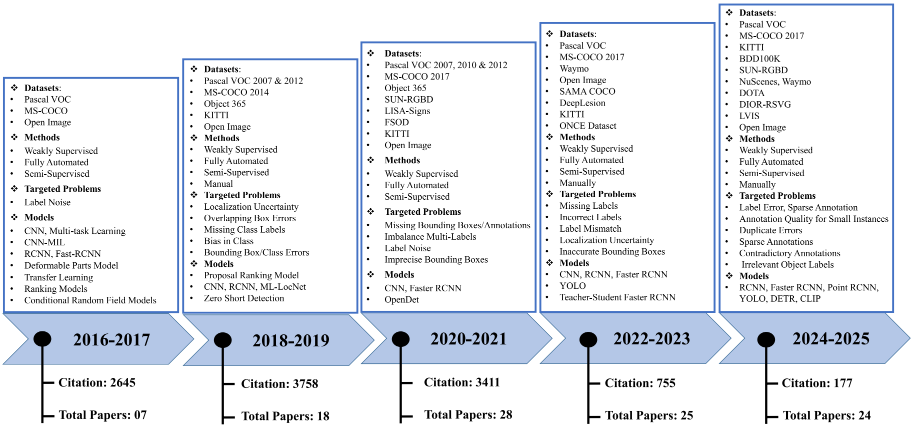

Fig. 2 Year-wise trend followed in the DVLPE domain, including a number of published papers and their citations in specific years (statistics covered from 2016 to 2025)

datasets. Furthermore, the trend has increased over time from 2022 to 2025, as more errors in the OD datasets are explored, and different semi-supervised, weakly supervised, and fully supervised methods are introduced in the literature. Although these methods address some problems in the datasets, there is still significant room for improvement.

3 Methodology

This section provides a detailed overview of the selection criteria of the datasets and literature on the detection and validation of label errors. In the literature, various techniques are presented for such a domain, and each method follows a generic process with some specific steps. We consider two major sources of errors in OD, such as dataset errors and model-driven errors. Dataset-driven errors refer to inaccuracies in the dataset itself, while model-driven errors refer to mistakes made by the model during the detection or classification process. Four different techniques are used for detecting annotation errors in OD datasets and model prediction, including manual error identification, semi-supervised, weakly-supervised, and fully automatic methods, as shown in Fig. 3. These methods are elaborated individually in the subsequent sections, and a comprehensive overview is shown in Table 3. It categorizes the surveyed work on the basis of evaluation metrics, datasets, publication year, and methodological category, which includes manual, weakly supervised, semi-supervised, and fully automatic methods. The review underscores the development of research from manually selected methods to semi-supervised and completely automated systems that prioritize scalability and reduce human involvement. Recent research has shown a significant movement toward automation and generalization in increasing annotation quality, with a focus on model-driven and self-learning frameworks used on large-scale benchmarks such as MS-COCO and PASCAL VOC.

3.1 Dataset and literature selection criteria

We selected datasets and relevant literature based on specific criteria to create a comprehensive and useful survey. We focused on open-source datasets that had a substantial impact on model development and evaluation. Benchmark datasets like MS-COCO (Lin et al. 2014) and its variants, PASCAL VOC (Everingham et al. 2013), ImageNet (Russakovsky et al.

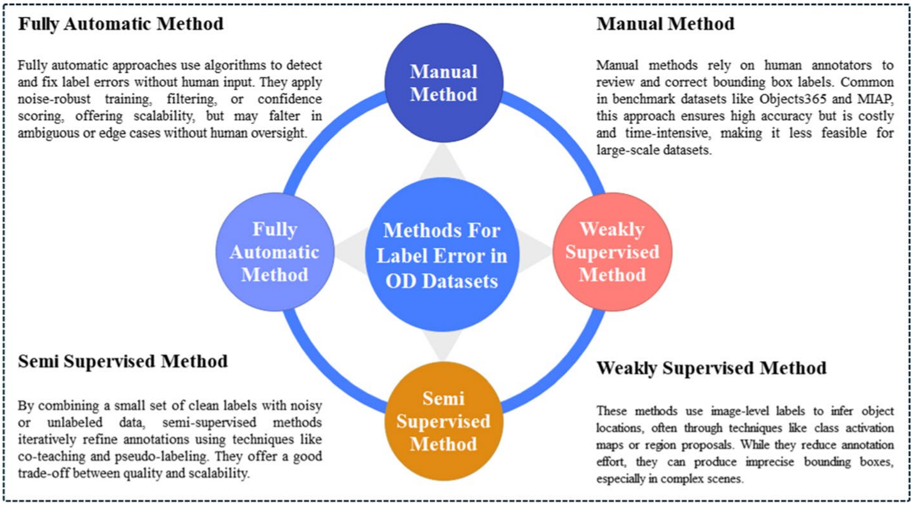

Fig. 3 Overview of the various strategies that are applied to address the annotation errors in object detection datasets. The illustration shows four of the most common techniques that are used to overcome inconsistencies in annotations and improve the accuracy of bounding boxes

2015), open images (Kuznetsova et al. 2020), Object365 (Shao et al. 2019), FSOD (Kang et al. 2019), DOTA (Ding et al. 2021), and KITTI (Geiger et al. 2012), are widely used in the field to train and validate detection systems.

Moreover, this survey offers an extensive overview of the challenges and approaches associated with errors in OD datasets. To ensure the intended level of relevance and quality of the included papers, the following inclusion and exclusion criteria are used. For the inclusion, we selected those papers that discuss or analyze some aspects of annotation error in object detection tasks. Studies that deal with the detection or validation of errors, such as mislabels, localization errors, or missing annotations, were prioritized. Published papers from peer-reviewed journals and select conferences deemed to be among the top in the field (i.e., IEEE, Springer, Elsevier, CVPR, ICCV) were included. Similarly, for the exclusion, we exclude those papers that are not about object detection tasks, or those that primarily focus on classification, semantic segmentation, or any task other than detection. Papers that did not provide a significant advancement in the detection or validation of annotation errors in datasets for object detection are also omitted.

Furthermore, this review specifically focuses on annotation errors in datasets used for object detection, so we include papers that are dedicated to the topic. Some of the related and key literature on annotation errors in datasets has been published and presented at top conferences—CVPR, ICCV, and ECCV, given the rapid pace of development in annotations. Top conferences tend to have cutting-edge, novel research, much of which lays the groundwork for future journal publications. We included conference papers primarily because they often provide new or novel findings or methods for detecting and/or correcting annotation errors, which is important to our survey. That's why we consider both journals and top conference papers for our survey.

3.2 Manual method

Improving dataset quality through manual methods involves processes where human annotators verify and revise annotations. These annotators identify inaccuracies in labeling, such as misclassified objects, missing objects, and bounding boxes that are either too loose or too tight. The annotation process generally follows established standards and includes multiple checks to ensure consistency and accuracy. Humans have the interpretive skill to correct labels effectively, particularly in cases involving occlusions, crowded scenes, and visually ambiguous objects (Schumann et al. 2021; Lin et al. 2014; Ma et al. 2022).

Manual annotation frameworks are utilized effectively by different large-scale datasets, such as Objects365 (Shao et al. 2019) and MIAP (Schumann et al. 2021). Objects365 followed a multi-stage annotation pipeline with initial annotation, validation, and refinement, all performed manually, generating over 10 million fine-grained bounding box annotations for 600 object classes (Shao et al. 2019). In the same way, the MIAP dataset, a comprehensive extension of Open Images, used extensive human labeling to annotate all individuals visible in each image and captured visually observable attributes, e.g., apparent gender and age. Interestingly, the study discovered that approximately 30% of individuals in the images were not tagged in the original dataset, underscoring the limitations of automated tools and the importance of human review (Schumann et al. 2021). Additionally, the high-risk nature of these diagnostic tasks means that manual verification is still essential in clinical practice (medical imaging). In the LUNA16 (Setio et al. 2017) dataset, the authors implemented multi-reader validation, where four radiologists confirmed each nodule. Likewise, the VinDr-Mammo (Nguyen et al. 2023) project employed dual-radiologist consensus to reduce labeling inconsistency and omitted-lesion errors. Such practices resemble manual error inspection protocols performed on general-domain datasets like Objects365 and MIAP.

While offering high-quality labels, manual methods still suffer from issues related to scalability and cost-effectiveness. It is unrealistic to use such manual methods on some datasets with millions of samples, or on datasets that require periodic updates. Moreover, human fatigue, variability of interpretation, and differences in the behaviors of annotators introduce additional errors, resulting in the need for additional quality control requirements and consensus decision-making to achieve the required levels of accuracy and reliability (Vondrick et al. 2013; Russakovsky et al. 2015). For these reasons, many recent annotation projects offer a mix of manual and automated/semi-automated methods to improve the balance between accuracy and scalability (Kuznetsova et al. 2020).

3.3 Weakly-supervised method

Weakly-supervised learning (WSL) provides an attractive alternative to OD, especially when annotations like bounding boxes are costly or unavailable. WSL algorithms rely on weak signals (such as image-level tags); the only information they have is whether a certain object exists in the image without its exact location. These methods focus on dataset-related errors and use few annotations, often based on less precise or incomplete labels (like image-level tags) to identify and fix mislabels, missing annotations, or inconsistencies. Additionally, these algorithms aim to reduce model-driven errors by enhancing object localization and recognition with such weak labels. As a result, the model's performance improves on new data, even with little supervision. The overall workflow of the WSL process is shown in Fig. 4 and is structured in three primary stages. In *Stage*1, input images are first passed through a convolutional backbone and global pooling to form class activation maps (CAMs), which indicate the most distinctive regions of an image. These CAMs are then passed on to *Stage*2, which uses weakly supervised segmentation to refine these activation regions and generate coarse localization cues. *Stage*3 includes a multiple instance learning (MIL) layer that correlates object proposals with the global image label to determine bounding boxes and final detections. This pipeline significantly reduces the cost of human labeling by enabling object localization and identification without the need for pixel-level or bounding-box annotations.

The main problem is to train models to obtain localization information from such broad labels, particularly in complex situations with many objects or distractors. Wang et al. (2022) introduced a correspondence learning framework that aligns object-level features under weak supervision, reducing dependency on paired annotations and improving spatial consistency. Subsequent studies (Diba et al. 2017; Bilen and Vedaldi 2016) further enhanced localization accuracy through cascaded learning and multiple instance reasoning. Many of these methods build on classification models, particularly convolutional neural networks (CNNs), which tend to highlight the most discriminative regions in an image. Techniques like class activation mapping (CAM) tap into this behavior to produce heatmaps, which loosely identify areas responsible for specific class predictions (Zhou et al. 2016). However, such maps frequently emphasize only parts of the object, such as a dog's face or a car's headlights, rather than outlining the entire instance. To overcome this limitation, researchers have developed techniques such as adversarial erasing and region-growing mechanisms

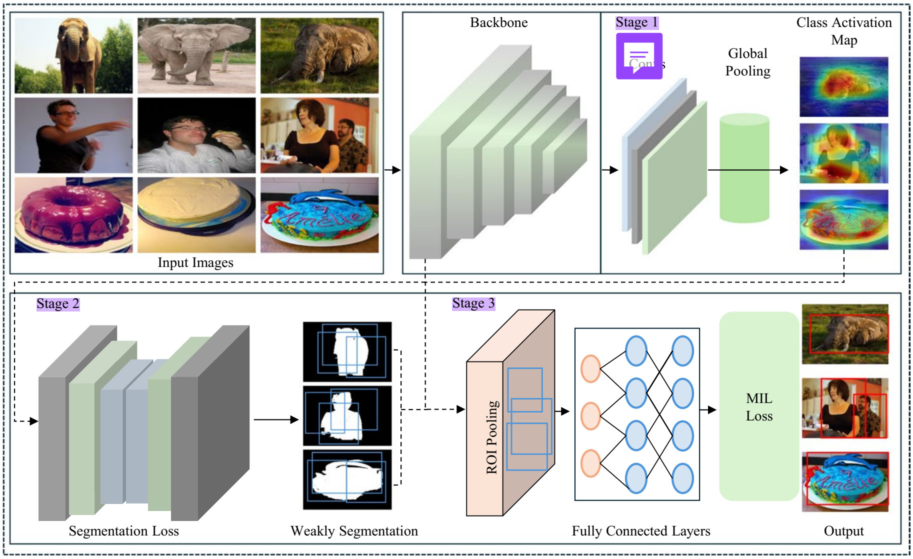

Fig. 4 Overview of the weakly-supervised method (Diba et al. 2017)

to broaden the highlighted areas and increase localization quality (Wei et al. 2017; Zhang et al. 2018).

Furthermore, one particularly important addition is SILCO (show a few images, localize the common object) (Hu et al. 2019). Unlike single-image approaches, it utilizes a collection of weakly labelled images and detects shared visual signals to determine the most likely object locations. Even with a small number of labelled samples, the cross-image consistency technique works well. Similarly, Weakly Supervised Deep Detection Networks (WSDDN) (Bilen and Vedaldi 2016) address OD with a multiple instance learning (MIL) approach in which the model selects proposals that best fit with the global image-level label without relying on annotated bounding boxes.

Recently, the performance of WSL approaches has been improved by using supplementary cues. Visual saliency, semantic segmentation masks, and even contrastive learning objectives all contribute to the model's ability to focus more correctly on object boundaries and reduce false detections (Ahn and Kwak 2018). Some approaches include self-supervised learning components or employ weak supervision as a preliminary phase before fine-tuning with stronger labels (Zhang et al. 2023; Hou et al. 2021). While WSL models may not yet be as precise as their fully supervised equivalents, they are critical in early dataset building, basic annotation production, and scalable detection algorithms. Their capacity to generate meaningful visual signals with minimum supervision makes them extremely useful, particularly in scenarios involving under-resourced domains or large-scale uncurated datasets.

3.4 Semi-supervised method

In recent years, semi-supervised learning (SSL) strategies have gained significant attention as a practical alternative to fully supervised training for object detection (OD), especially when the cost or feasibility of dense annotations, like bounding boxes, is a limiting factor.

These techniques rely on incorporating a smaller number of labeled samples along with a large amount of unlabeled data, allowing models to generalize beyond the data that has been annotated. Such methods intend to correct errors arising from datasets and reduce annotation errors (e.g., mislabeling or lack of bounding box) using a small number of labeled samples. In improving the model's robustness to predict accurately despite incorrect ones or for sparse annotations, they are also correcting the prediction errors that arise from the model. To avoid explicit manual annotation and also maintain performance benefit by such a strategy, it is essential to design mechanisms such as pseudo-label generation, consistency enforcement, and teacher-student learning dynamics. The general flow of the SSL framework is presented in Fig. 5, which uses labeled and unlabeled data. In particular, the high-level features of both support and query datasets are first extracted by a backbone model. A teacher-student analogy is held between these two branches. Spatial similarity and feature sample modules guide the learning process to share reliable knowledge from labeled samples to unlabeled ones. It tries to improve the pseudo-labels in the unlabeled set iteratively, and thus achieves better detection performance and generalization. This framework balances the model's performance and annotation efficiency by using limited supervision with large amounts of unlabeled data.

Focusing on the earliest research direction, which mainly aimed to make detection models robust to incorrect labels. Such as Vahdat (2017) proposed a technique that leveraged robustness-aware training in the presence of noise to ensure the model was effective even in the presence of invalid annotations. Choi et al. (2018) introduced a different approach in which the training loop was incorporated with statistical co-occurrence information so that the model could receive indirect supervision. In addition, Zhang et al. (2018) illustrated the advantages of partially labelled datasets employing adaptive learning signals through weak supervision. Jeong et al. (2019) enhanced generalization by means of consistency regularization, which encourages the model to produce confident predictions in the presence of variations.

Many of the subsequent advancements drew inspiration from pseudo-labeling principles, especially through teacher-student paradigms. In these frameworks, a stronger teacher network generates supervision signals for a student model trained on unlabeled data. Examples include the instant-teaching framework (Zhou et al. 2021), which incrementally changes pseudo-labels, and the virtual category learning framework (Chen et al. 2022), which uses structured category-level feedback to improve semantic alignment. Soft teacher model (Sohn et al. 2020) also applies threshold-based filtering to improve precision on the pseudo-

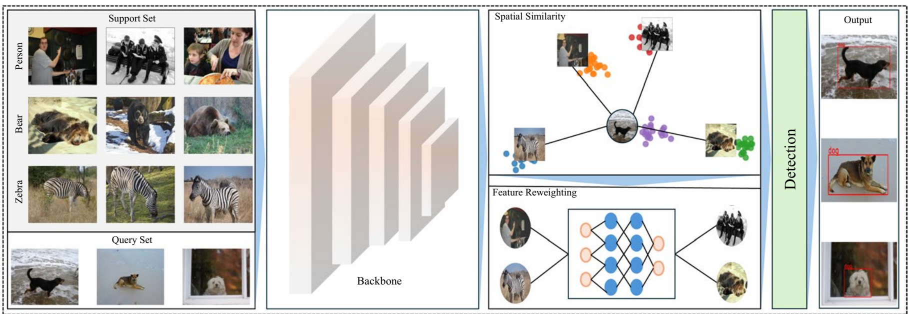

Fig. 5 Overview of the semi-supervised method (Hu et al. 2019)

labels. Some of the ideas have been transferred to 3D OD tasks, especially those involving LiDAR and robotics, where, for instance, reliable student (Nozarian et al. 2023), and successors (Chen et al. 2023; Han et al. 2024), and Liu et al. (2024) use the cohesion of temporal data, multi-view reasoning to minimize point cloud prediction errors. SSL methods handled high variability with great ease in remote sensing and aerial imagery. Kim et al. (2025) also proved that noisy bounding boxes are still useful signals if noise-robust loss functions are applied. In remote sensing and aerial imagery, several works have applied semi-supervised learning. Liu et al. (2024) proposed the model to introduce a paradigm of uncurated, unlabeled data, which iteratively refines pseudo-labels to improve robustness to scale and orientation variations. Other works (Cai et al. 2022; Choi et al. 2022; Zhao et al. 2025) that have encountered the same problem affirmed that complexity in the visual domain is proportional to the degree of spatial cohesion and selective updating of pseudo-labels. Also, medical object detection SSL methods were used to address the absence of bounding boxes. The Lesion Harvester (Cai et al. 2020) proposed pseudo-label mining for the DeepLesion dataset (Yan et al. 2017), where the unlabeled lesions are automatically discovered and iteratively refined with teacher-student retraining. Likewise, Bai and Xia (2022) proposed a universal lesion detector that jointly learns from labeled and unlabeled CTs, explicitly correcting annotation omissions. Such methods exemplify the use of semi-supervised strategies for dataset-error remediation, paralleling the general SSOD approaches summarized earlier.

Several approaches have emerged for directly managing pseudo-label quality and uncertainty. Gao et al. (2019) presented ensemble refinement to reduce false positives, whereas Wang et al. (2021) proposed region-wise confidence estimation to detect and rectify inaccurate predictions before they affect training. Further advances, such as WSSOD (Fang et al. 2021), combined weak and semi-supervised techniques into unified pipelines, using weak labels as anchors to increase supervision. Other research, including (Zhang et al. 2019; Guo et al. 2021), and Ryoo et al. (2023), investigated the concept of mining latent annotations, which are things that were overlooked during manual labelling but can be recovered via unsupervised processes.

To tackle the challenging training environments, open-set recognition and long-tail distribution models were suggested by researchers. Models that dynamically adapt to new or rare categories of objects were built in Joseph et al. (2021), Huang et al. (2022), and Zhong et al. (2024), assessing new detection extensions using contrastive attributes and category extensions, and extending detection capabilities. While others Zhao et al. (2020) highlighted the necessity to harmonize label spaces across datasets for domain-independent object detectors. Data sparsity, which is another major challenge, has been helped by advances such as inter-model communication (Dong et al. 2018), data-efficient fine-tuning (Xiong et al. 2021), and student ensemble training (Wu et al. 2024). To fill the gaps between minimal supervision and extensive annotation, domain-specific priors were considered in the detection. Hu et al. (2019) introduced SILCO, which describes the models to capture consistent properties of the objects in distinct images, which assist in the inference of bounding boxes in the absence of any explicit labels. Another study by Schumann et al. (2021) enhanced ethical proposals for annotation by ensuring detection systems capture a more diverse sampling of the human. Also, the methods provided in Zhou et al. (2021), Chen et al. (2022), Chen et al. (2022), and Liu et al. (2022) used adaptive matching and open set modelling, providing solutions to label mismatch and unknown class discovery.

Further advanced solutions have been developed as the field has evolved. which consists of techniques such as geometric alignment (He et al. 2021), dense pseudo-label aggregation (Yang et al. 2025), and semi-DETR semi-DETR (Shehzadi et al. 2024), which shows attention-guided sparse query learning. Some researchers have extended SSOD methods to specific domains such as agriculture and disaster management, where models are required to perform with little or no labeled data. The work such as Liu et al. (2024), Wang et al. (2025), Zhao et al. (2025), and Zhu et al. (2025) demonstrates how consistency learning, ensemble feedback and region refinement can be employed in real-world impact domains. Furthermore, the automated data labeling tools that combine active learning with adaptive data augmentation (Vijiyakumar et al. 2024) are a positive development towards the SSOD frameworks.

In summary, the research of semi-supervised learning now incorporates various techniques and applications. Common to all of these, we identify innovations in strategies for refining pseudo-labels, handling imperfect data, adapting them dynamically as training progresses, and generalizing across domains. Together, these contributions demonstrate that SSOD is an efficient, scalable, and cost-effective alternative to traditional annotation-heavy pipelines in OD research and deployment..

3.5 Fully automatic method

There is a growing need for automatic OD and label-correction methods due to the exponential increase in large visual datasets. These systems have algorithmic intelligence that detects objects, corrects labeling mistakes, and reduces annotation noise, making it possible for fully automatic methods (FAM) to work in a human-free control way. They consider both model-driven mistakes (missed and false detections) and dataset-driven challenges (mislabeled examples, wrong boxes). FAM has particularly strong performance on large, intricate datasets, as illustrated in Fig. 6. The workflow begins with raw images being processed by a backbone and a region proposal network (RPN), which generates proposed object regions. Confidence-based filtering inputs the refined boxes over shared layers and ROI mapping; the final detection module (classifier, regressor, and estimator) generates stable noise-robust predictions. The method automatically identifies and corrects annotation mistakes, so it is applicable for large and continuously updated datasets.

FAM systems have proven to be valuable in scenarios involving large, diverse data where manual analysis is not possible and for real-time systems that need self-adjustment. Preliminary work in this direction emphasized robust detection under noisy visual environments. For instance, Milyaev and Laptev (2017) presented adaptive thresholding techniques to enhance detection performance in the presence of degraded image conditions. The problem of weakly annotated training scenes was also dealt with through rank-based transfer learning (Shi et al. 2017) to enable models to automatically discover and rank high-quality candidate regions. Exploiting this, researchers proposed models to process web-scale noisy datasets via loss-based filtering, as done in Veit et al. (2017) or click-based bounding box estimation (Papadopoulos et al. 2017), reducing annotation effort by extrapolating box coordinates from single point annotations.

Advances in bounding box refinement approaches have enabled models to modify spatial parameters without explicit supervision. Recent research has focused on achieving precise localization in noisy environments. Regression-sensitive suppression techniques improved

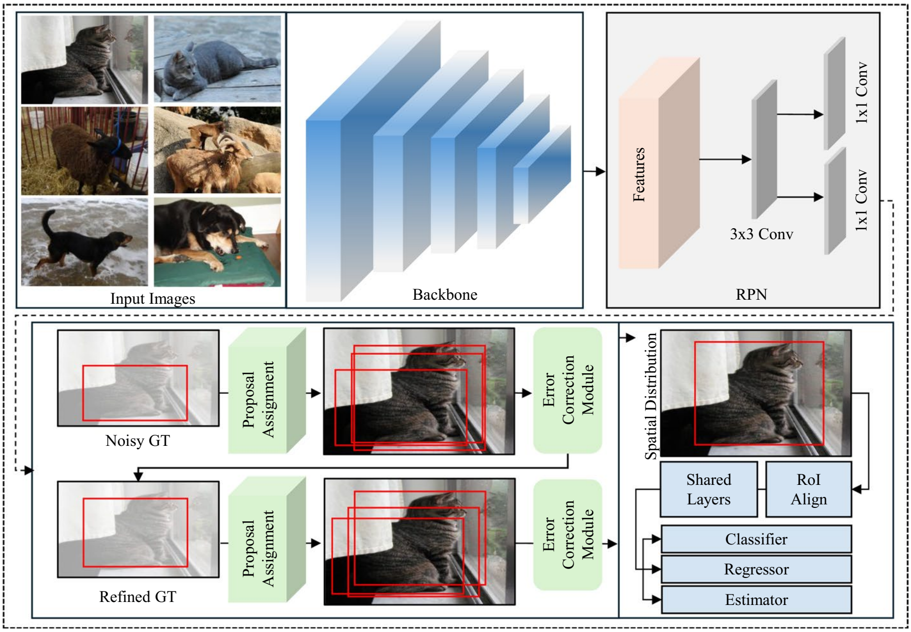

Fig. 6 Overview of the fully automatic method (Zhou et al. 2023)

spatial alignment during post-processing by considering the confidence of predicted boxes (He et al. 2018). Multiview learning frameworks were also developed to improve object localisation by combining complementary viewpoints (Zhang et al. 2018). Simultaneously, methods such as zero-shot detection (Bansal et al. 2018) and low-shot transfer learning (Chen et al. 2018) enabled models to recognize novel or under-represented categories, increasing the generalizability of detection frameworks. Adversarial training techniques (Zhang and Wang 2019) and proposal re-ranking procedures (Tan et al. 2019) were used to give priority to high-quality inputs. These methods provided resistance against input uncertainty and label fluctuations. Distribution-aware sampling (Wu et al. 2018) and robust soft sampling approaches (Chadwick and Newman 2019) increased training reliability in scenarios with highly imbalanced data or partial annotations.

Later research focused on learning from synthetic labels or poor bounding boxes. Datasets with unclear label quality (Li et al. 2020), affected images (Nayan et al. 2020), and noisy anchor points (Li et al. 2020) were used to train the models. The decision boundary between the foreground and background classes was modified by a number of frameworks that used background recalibration losses (Shen et al. 2019; Zhang et al. 2020; Peng et al. 2020) to handle label sparsity and annotation omission. Others enforced consistent detection in weakly supervised settings by enforcing category-aware spatial constraints (Zhang et al. 2020). As the challenge of achieving complete annotations persists, positive-unlabeled learning paradigms (Dhamija et al. 2020), open-set detection frameworks (Yang et al. 2020), and anchor-free bounding box refinement (Samet et al. 2020) have come to the forefront. These models effectively distinguish valid object instances even when class definitions are vague or incomplete. Bayesian modeling (Khanzhina et al. 2021) and contrastive learning

(Sun et al. 2021) were also employed to quantify uncertainty and improve few-shot detection under limited supervision.

The training dynamics of detectors in noisy environments have been enhanced by recent developments in loss formulations, such as feature-guided refinement targets (Zhang et al. 2021) and scale-balanced loss (Shuang et al. 2021). Methods that use robust regression objectives (Zhang et al. 2020; Chachula et al. 2022), and bounding box confidence reweighting (Xu et al. 2021) enable the dynamic suppression of poor predictions during learning. Temporal consistency enforcement is a recent innovation in which label propagation across video frames facilitates tracking and correction (Miller et al. 2021; Liu et al. 2022; Wu et al. 2023). In order to improve the internal consistency of models, self-distillation and spatial alignment modules (Chachula et al. 2022) were added. Additionally, studies used dynamic calibration techniques and uncertainty estimation to overcome open-set detection problems (Tkachenko et al. 2023; Bär et al. 2023; Zhou et al. 2023).

The need for benchmarking automated label quality motivated the creation of new datasets and diagnostic pipelines (Ryoo et al. 2023; Zimmermann et al. 2023; Pičuljan and Car 2023). Quality assurance systems were proposed to assess annotation validity using model-driven or data-driven inference, which has been applied to standard benchmarks like MS-COCO and PASCAL VOC (Tong and Wu 2023; Duan et al. 2023; Borji xxx). Cross-domain annotation transfer (Hosoya et al. 2024) and consistency-enforcing losses (Freire et al. 2024) further reduced manual re-labeling efforts. In real-world scenarios, models trained with uncertain lidar and camera input benefited from lightweight prediction quality modules (Suri et al. 2023) and semi-automated convergence analysis (Liao et al. 2024). Interpretable debugging tools (Riedlinger et al. 2025) were developed to locate problematic annotation slices, and label alignment pipelines allowed for automated resolution of contradictory inputs (Tschirschwitz and Rodehorst 2025). Recent advances in label pruning (Chen et al. 2025) and large-scale error discovery frameworks like ClipGrader have used vision-language models to evaluate the semantic consistency of annotations without the need for ground-truth validation (Kim et al. 2025; Lu et al. 2025). These changes are the end of a trend toward entirely autonomous detection of objects. This will let models learn, evaluate, and get better on their own, even in complex and evolving visual domains.

The capabilities of object detection (OD) and localization have been enhanced by recent developments in computer vision (Roy et al. 2023; Kumar et al. 2024; Khan et al. 2023; Singh et al. 2023), highlighting the necessity of reliable, data-driven methods. This development has been expanded upon by transformer-based architectures like DETR, DINO, RT-DETR, etc., which significantly improve object localization accuracy through attention-driven, query-based learning. Additionally, multimodal alignment of textual and visual representations is used by vision-language models *(VLMs) like CLIP, Grounding DINO, and CLIPGrader (Lu et al. 2025) to automatically identify mislabeled annotations and semantic variations. These frameworks, powered by transformers and VLMs, provide a new generation of completely autonomous, data-centric techniques that improve scalability and semantic awareness during dataset validation and refinement. These frameworks, powered by transformers and VLMs, provide a new generation of completely autonomous, data-centric techniques that improve scalability and semantic awareness during dataset validation and refinement.

In addition to annotation errors, image-level quality challenges like as blurriness, noise, and resolution deterioration influence object identification performance and can increase label inaccuracies. Recent research has demonstrated that deep transformer-based restoration networks, such as MC-Blur (Zhang et al. 2023) , DBLRNet (Zhang et al. 2018), and MB-TaylorFormer v2 (Jin et al. 2025), may restore fine-grained information and reduce the transmission of low-quality distortions throughout annotation pipelines. Cleaner and more dependable object-recognition datasets may be developed by integrating such image-enhancement frameworks with annotation-error detection methods.

To provide readers with a more in-depth understanding of the practical use and viability of the reviewed approaches, Table 2 outlines the actual workflow, use case scenarios, and pros and cons of each method category. This aims to provide readers with a clearer understanding of the practical applicability and trade-offs among the reviewed approaches. Subsequently, the Fig. 7 illustrates the quantitative comparison of the four methods in terms of six evaluation criteria, such as annotation accuracy, scalability, automation, cost efficiency, human effort (inverse), and adaptability. The figure visually highlights the trade-offs between accuracy, scalability, and automation. Manual methods achieve the highest annotation precision but require maximum effort, whereas fully automatic methods excel in scalability and cost efficiency with reduced human supervision. Weakly and semi-supervised approaches provide a balanced compromise between accuracy and efficiency.

Table 2 Comparison of manual, weakly supervised (WSL), semi-supervised (SSL), and fully automatic (FAM) methods for handling annotation errors in object detection datasets

Method	Workflow summary	Typical application scenarios	Advantages	Limitations
Manual	Human annotators visually inspect and correct bounding boxes and class labels through iterative verification and consensus steps	Small- to medium-scale datasets, or high-stakes applications (e.g., medical imaging, autonomous driving) where maximum label accuracy is required	Highest annotation precision and interpretability; capable of resolving ambiguous or complex visual instances	Time-consuming and labor-intensive; not scalable to large datasets; subject to human fatigue and variability
Weakly supervised (WSL)	Models trained with weak labels (e.g., image-level tags) infer approximate object locations using class activation maps or saliency cues	Suitable when bounding-box annotations are unavailable or costly; useful in large unlabeled or low-resource datasets	Reduces annotation cost; enables dataset expansion with minimal supervision; useful for early-stage dataset creation	Lower localization precision; often detects only salient object parts; requires additional refinement or post-processing
Semi-supervised (SSL)	Combines a small labeled subset with a large unlabeled pool using pseudo-labeling, consistency regularization, and teacher–student frameworks	Effective when limited annotations exist; applicable to scalable dataset building with moderate supervision effort	Balances annotation cost and performance; adaptable to multiple domains; improves model robustness to label noise	Performance sensitive to pseudo-label errors; requires careful confidence thresholding and consistency tuning
Fully automatic (FAM)	Automatically detects and corrects annotation errors using model-driven inference, confidence scoring, or self-learning without human input	Applicable to very large or continuously updated datasets; autonomous pipelines and large-scale annotation systems	Highly scalable and efficient; suitable for real-time or continuous dataset refinement; minimizes human intervention	Can introduce false corrections; lacks human interpretability; overall quality depends on the model's reliability

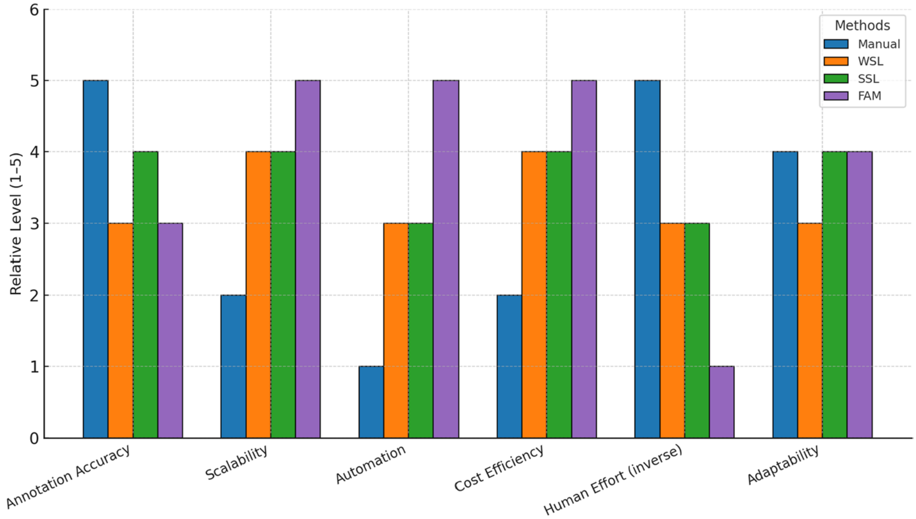

Fig. 7 Quantitative comparison of the manual, WSL, SSL, and FAM methods across six evaluation dimensions: annotation accuracy, scalability, automation, cost efficiency, human effort (inverse), and adaptability

3.6 Annotation toolchains and platform limitations

The tools and platforms utilized for annotation during the creation of OD datasets have a significant impact on the final dataset's quality. Although many tools are designed to expedite and simplify annotation, they may unintentionally add biases or limitations that compromise the consistency and quality of the annotations, affecting the model's performance.

Most annotation tools, like Labelbox, VGG image annotator (VIA), and RectLabel, are built to assist users with the tedious task of annotating images and complex objects. However, features that enhance the usefulness of these tools can also lead users to make mistakes. For example, some tools automatically generate bounding boxes around a given object. While this is a helpful feature because it creates a bounding box in less time, it might create the bounding boxes incorrectly by being too tight, too loose, missing parts of the object, or including unnecessary background. Localization errors hinder model training because some tasks require precise representations of target objects. Additionally, systems often struggle with annotating ambiguous or difficult objects. Over-reliance on features like auto-generated boxes or visualization may cause annotators to make systematic errors. Research indicates that tools with limited support for occlusion or small objects often produce annotations that underestimate such challenging instances (Schumann et al. 2021). This affects the robustness of the model when these objects lead to poor training data (Zhou et al. 2016).

Moreover, human or annotator biases can also stem from the tools they use during annotation. Furthermore, human biases influence how annotators utilize annotation tools. They may accept auto-generated suggestions from tools based on prior work without thoroughly reviewing them. This results in a confirmation bias, in which the tool's suggestions affect the final annotations even when they are incorrect. Research shows that although auto-labeling

Table 3 Detailed description of the error detection and correction papers

Method	Year	Dataset	Det	EM	Description	GitHub
FAM(Bansal et al. 2018)	2018	MS-COCO, Visual Genome, OpenImages	Two Stage	IOU	Solves zero-shot category generalization, unseen-as-background mislabeling, seen-class dominance, and background class homogenization errors	–
SSL (Chen et al. 2018)	2018	MS COCO, ImageNet, PASCAL-VOC	Two Stage	mAP	Solves overfitting under scarce labels, transfer mismatch, background over activation, underutilized source knowledge, and cold start detection errors	GitHub
WSL (Hu et al. 2019)	2019	PASCAL-VOC	One Stage	mAP	Solves box annotation absence, weak label localization gap, support sample noise, saliency-driven localization error, and small object localization error	–
Manual (Shao et al. 2019)	2019	MS-COCO, PASCAL-VOC, Objects365	Both	mAP, IOU, MR	Solves incomplete annotation, low label recall, bounding box inconsistency, label ambiguity, iconic image bias, partial network initialization, and training inefficiency	–
SSL (Zhang et al. 2020a)	2020	MS-COCO, PASCAL-VOC	Both	mAP, CorLoc	Solves missing annotation, partial OD, pseudo ground truth noise, annotation strategy bias, and detector degradation under sparse supervision	–
FAM (Li et al. 2020)	2020	MS-COCO, PASCAL-VOC	Two Stage	mAP	Solves class label noise, bounding box coordinate noise, entangled annotation noise, false positive annotation error, and confirmation bias	–
FAM (Zhang et al. 2020b)	2020	MS-COCO, PASCAL-VOC	One Stage	mAP	Solves missing ground truth, confusing anchor signal, hard negative gradient spike, training instability under partial labels, and sampling bias under sparse labels	–
FAM (Dhamija et al. 2020)	2020	MS-COCO, PASCAL-VOC	Both	mAP, AWI	Solves unknown object misclassification, mixed unknowns error, undiagnosed open-set error, overconfident objectness error, and wilderness vulnerability	GitHub
FAM (Shuang et al. 2021)	2021	MS-COCO, PASCAL-VOC	Both	mAP,AR, Recall	Solves anchor matching imbalance, small object underrepresentation, intra-class foreground bias, redundant anchor contribution, and gradient dominance instability	GitHub
FAM (Joseph et al. 2021)	2021	MS-COCO, PASCAL-VOC	Two Stage	mAP	Solves open-set misclassification, unknown-background confusion, incremental forgetting error, false positives from unknowns, and closed-set evaluation bias	–
Manual (Schumann et al. 2021)	2021	Open-Images	_	Box Count, nPMI	Solves missing person annotation, subclass label supremacy, stereotype-driven annotation, classifier proposal bias, and demographic label incompleteness	–

Table 3 (continued)

Method	Year	Dataset	Det	EM	Description	GitHub
SSL (Xu et al. 2021)	2021	MS-COCO, PASCAL-VOC	Both	mAP, AP, APS, APM, APL	Solves label noise error, localization noise, class-box mismatch, training instability, and over-pruning of informative samples	–
SSL (Chen et al. 2022; Klie et al. 2023)	2022	MS-COCO, PASCAL-VOC	Two Stage	mAP	Fixes confirmation bias and confusing samples by using a virtual category, boosting accuracy with few labels	GitHub
FAM (Miller et al. 2022)	2022	MS-COCO, PASCAL-VOC	Both	mAP, AUROC, TPR	Solves open-set errors where unknown objects are wrongly detected as known classes with high confidence, by using uncertainty estimation to identify and reject these false positives	GitHub
SSL (Cai et al. 2022)	2022	MS-COCO, PASCAL-VOC	Two Stage	mAP, AP	Solves pseudo label localization error and pseudo label training noise by using strong-weak heads in a teacher-student model, ensuring better label filtering and noise-tolerant training	–
SSL (Chen et al. 2022)	2022	MS-COCO, PASCAL-VOC	One Stage	mAP	Fixes pseudo-label noise, false positives, teacher instability, and uncertainty inconsistency in anchor-free SSOD using adaptive filtering, class-aware refinement	–
Manual (Ma et al. 2022)	2022	MS-COCO, Open-Images	Both	mAP, AP, APS	Fixes annotation noise, bounding box inconsistency, semantic ambiguity, and cross-dataset annotation policy errors through expert-guided reannotation	–
FAM (Liu et al. 2022)	2022	MS-COCO, PASCAL-VOC	Both	mAP	Solves bounding box noise, instance selection noise, and low-quality proposal errors using an object-aware multiple instance learning	GitHub
FAM (Wu et al. 2023)	2023	MS-COCO, PASCAL-VOC, Objects365	Both	AP, mAP	Solves inaccurate bounding box, object drift, group prediction, and part domination errors using spatial self-distillation to enhance robust box refinement and proposal selection	GitHub
FAM (Chachuła et al. 2023)	2023	MS-COCO, Waymo, Sama-COCO	Both	mAP	Fixes noisy annotation errors like missing, spurious, mislocated, and mislabeled boxes in OD datasets, boosting model performance through data cleaning	–
FAM (Tkachenko et al. 2023)	2023	MS-COCO, SYNTHIA	Two Stage	mAP	Solves overlooked bounding box, badly located box, swapped class label, and annotation inconsistency errors by automatically scoring and prioritizing mislabeled images	GitHub
FAM (Ryoo et al. 2023)	2023	MS-COCO, Open-Images	Both	mAP, AP, TIDE	Defines and analyzes categorization noise, localization noise, missing annotation, and bogus bounding box noise	GitHub

Table 3 (continued)

Method	Year	Dataset	Det	EM	Description	GitHub
FAM (Pičuljan and Car 2023)	2023	MS-COCO	_	std, Precision, Recall	Solves bounding box quality errors, including label distortion, deletion, class swaps, and bogus labels	–
Manual (Tong and Wu 2023)	2023	MS-COCO, PASCAL-VOC, Mini6KC, Mini6K, Mini2022, SDOD	Both	mAP, APM, APL, Precision, Recall, F1	Solves missing annotation, class label noise, bounding box misuse, and duplicate annotation errors	–
FAM (Suri et al. 2023)	2023	MS-COCO, PASCAL-VOC	Two Stage	AP	Solves false negatives from missing labels, noisy pseudo-label degradation, classifier over-penalization, and benchmark inconsistency	GitHub
FAM (Schubert et al. 2024)	2024	MS-COCO, PASCAL-VOC, BDD100K, KITTI	Two Stage	AUROC, F1, Precision	Solves missing label, class label error, background false positive, and bounding box misalignment by analyzing per-instance losses	GitHub
SSL (Wu et al. 2024)	2024	MS-COCO, BDD100K, TUPAC16	One Stage	AP, F1	Solves unlabeled foreground error, pseudo-label noise, augmentation-induced mislabeling, and inter-student label conflict	GitHub
FAM (Hosoya et al. 2024)	2024	MS-COCO	Both	AP, APS, APM, APL	Solves small object annotation cost inefficiency, small object omission, domain gap from size-based scaling, and training data imbalance across object sizes	–
SSL (Shehzadi et al. 2024)	2024	MS-COCO, PASCAL-VOC	One Stage	mAP, APS, APM, APL	Solves inaccurate pseudo-label, duplicate prediction, small O, and inefficient query usage errors using query refinement and pseudo-label filtering	–
FAM (Duan et al. 2024)	2024	MS-COCO	One Stage	AP, APS, APM, APL, AR	Solves geometric bias, false bounding boxes, feature deficiency, central region mismatch, and latency bottlenecks in OD using a bottom-up keypoint triplet strategy	GitHub
FAM (Zhou et al. 2024)	2024	MS-COCO, PASCAL-VOC	Two Stage	AP,APM, APL, mAP	Solves noisy bounding box, corrupt supervision signal, prediction confidence ambiguity, and category feature bias	–
FAM (Liao et al. 2024)	2024	Waymo, Cityscapes, MVD, nuScenes, Synscapes	Both	mAP, AP	Solves class semantics mismatch, annotation instruction discrepancy, human–machine label misalignment, and cross-modality labeling error	–
SSL (Wang et al. 2025)	2025	MS-COCO, PASCAL-VOC	Two Stage	mAP, AP	Solves partial utilization of unlabeled data, data augmentation redundancy, class imbalance in sample weighting, and uncertainty metric bias	–
FAM (Kim et al. 2025)	2025	DOTA, DIOR-R	Both	mAP	Solves bounding box localization noise, annotation overfitting, tiny object perturbation, and aspect ratio distortion using a noise-injection strategy during training	GitHub

Table 3 (continued)

Method	Year	Dataset	Det	EM	Description	GitHub
FAM (Kim et al. 2025)	2025	KITTI, Waymo, nuScenes	Two Stage	AP	Solves misclassification, bounding box misalignment, annotation omission, label corruption, label confusion, class imbalance, and redundant data errors	–
FAM (Lu et al. 2025)	2025	MS-COCO, LVIS	_	Accuracy, FPR	Solves class label mismatch, bounding box localization error, background labeling error, pseudo-label noise, and annotation inconsistency	–
FAM (Kim et al. 2025)	2025	MJ-COCO, MS-COCO	Both	AP, AUROC, PRAUC	Solves Missing object labels, Incorrect class assignments, Inaccurate bounding boxes, Group labeling, Duplicate labeling and Incorrect object estimations	GitHub

EM evaluation metrics and *Det* detector

features are practical, they can lead to errors, particularly when the program learns wrong patterns from previous annotations (Wei et al. 2017).

Limitations in tool design can lead to bottlenecks in data management. Certain tools lack effective systems for handling complex categories or scenarios with multiple objects. For example, when several similar items are near together, tools may fail to differentiate among them, resulting in grouping errors. Such challenges are common in datasets such as MS-COCO, where objects often overlap and annotators must label many objects in a single image (Kuznetsova et al. 2020). Such errors have a significant impact on model performance, particularly for tasks requiring small or crowded objects. Finally, reducing tool biases aims to increase model reliability as well as annotation accuracy. Efficient tools provide better data, and better data is necessary to develop object detection model systems that are more efficient and broadly applicable.

3.7 Inspection and validation of annotation errors in object detection benchmarks

Understanding the reliability of benchmark datasets is a prerequisite for trustworthy model evaluation. Since annotation errors can lead to misleading conclusions about model performance, it is essential to assess the datasets themselves with the same rigor applied to models. To this end, we manually analyzed various high-impact datasets to identify and validate common error categories, as specified in Sect. 4. In particular, we examined DOTA, open images, Pascal VOC, FSOD, Object365, LVIS, and MS COCO, which are well-known datasets. We aimed to determine, classify, and measure the different types of errors in the annotations using the definitions described earlier. This stage required us to examine thousands of images manually to determine whether there were any systematic errors and to organize every instance of such errors into one of the predetermined categories. The summary of all errors in the dataset is provided in Table 7, and Fig. 8. All identified instances are documented with image ID in the Supplementary file.

From all available datasets, around 47, 000 images were examined. We manually checked them after performing random selection for images available in each of the datasets. For the specialized small-scale datasets like DOTA, we manually checked the entire dataset. For the

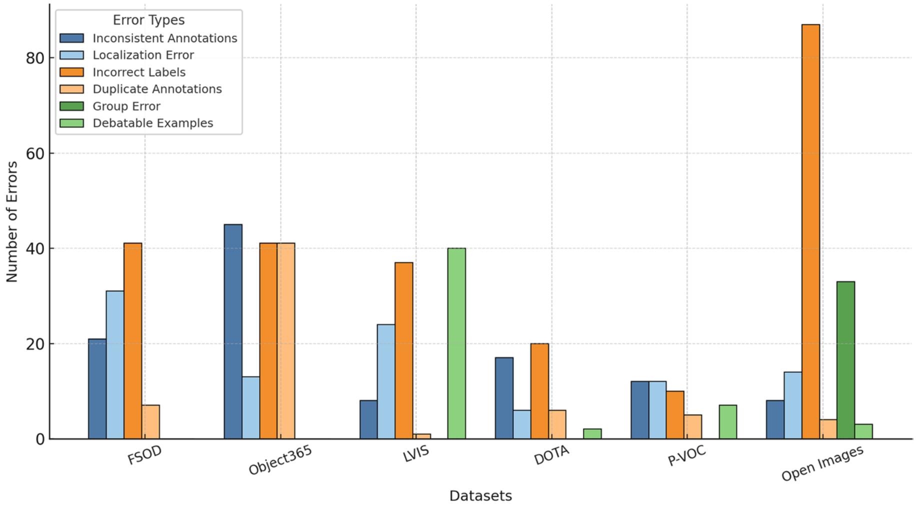

Fig. 8 Annotation errors across benchmark object-detection datasets, showing dominant incorrect label and localization errors in large datasets and a more balanced error distribution in smaller ones

larger general-purpose datasets like OpenImage, we performed random sampling from the entire dataset, but we made sure to cover all the dataset's classes. We followed a protocol where errors were classified into six categories: (1) Inconsistent annotations, (2) Localization errors, (3) Incorrect labels, (4) Duplicate annotations, (5) Group errors, and (6) Debatable errors. These errors are further discussed in the following section. Each dataset was visually checked using a Python script, and the images and errors were logged manually. For each dataset, the images and the errors are provided in the supplementary file.

We used an iterative validation approach with a team of 15 people divided into two subteams. In the first step, *Team* 1, which consisted of 10 individuals, used the Python script to identify possible errors (such as labeling and bounding box issues) in the dataset and record their information. These detected errors were subsequently passed on to *Team* 2, which consisted of five members. *Team* 2 verifies those errors. Following verification, the procedure was repeated, with the teams working together to identify and validate all the errors in the dataset. This iterative approach increases the precision and consistency of error detection because errors are cross-checked by many annotators, which improves the reliability of the final results.

3.7.1 Errors identified in FSOD

We analyzed the few-shot object detection (FSOD) dataset and identified 100 samples with different types of errors. The most prominent error identified among all the errors was labeling errors, which arose in 41 samples. The next most common error was a data localization error, where 31 samples were detected with this error. These types of errors are especially important for low-data regimes since such minor noise in datasets can greatly affect the generalization of the model. We also identified bounding box errors (21) along with annotation errors (7), as provided in Table 4, underscoring the need for more rigorous annotation

Table 4 Manually identified annotation errors across various benchmark datasets

#	Error Types	FSOD	Object365	LVIS	DOTA	P-VOC	Open Images
1	Inconsistent Annotations	21	45	8	17	12	8
2	Localization Error	31	13	24	6	12	14
3	Incorrect Labels	41	41	37	20	10	87
4	Duplicate Annotations	7	41	1	6	5	4
5	Group Error	-	-	-	-	-	33
6	Debatable Examples	-	-	40	2	7	3

protocols even for smaller datasets. All the errors with image IDs are provided in Supplementary file Table 1.

3.7.2 Errors Identified in Object365

To analyze the Object365 dataset, we identified a total of 140 samples with different types of annotation errors. The most common annotation error is an inconsistent annotation error, which arises in 45 samples. Similarly, the other type of annotation errors, such as duplicate bounding boxes (41), labeling errors (41), and localization errors (13), were also present. There is a higher annotation complexity when label density and scene clutter are present, which is why quality control within the annotations of big datasets like Object365 is especially important. All the annotation errors and IDs identified in the Object365 dataset are provided in the Supplementary Table 2.

3.7.3 Errors identified in LVIS

The LVIS dataset is a long-tailed OD benchmark; we assessed 25, 000 samples and identified 112 samples where annotation errors were present. Although LVIS demonstrated a strong labeling pipeline and concentrated on rare categories, various forms of errors were still identified, as provided in Supplementary Table 3. The most common forms of error were challenging and debatable annotations, which arose in 40 samples, as well as incorrect labels (37), many of which concerned visually similar or fine-grained categories. There were also reports of other errors, such as localization errors (24) and inconsistent annotations (8). Furthermore, rare cases of contradictory labeling (2) and duplicate annotation (1) are also observed, as provided in Table 4.

3.7.4 Errors identified in DOTA

From the aerial imagery-based DOTA dataset, we evaluated 1869 total samples and identified 51 erroneous samples. The most common error within the dataset was incorrect labels (20), followed by inconsistent annotations with 17 samples. Similarly, the same number of localization errors as well as duplication errors (6 each) were also identified. Such inconsistency was often due to occlusions, the number of objects, and the differences in scale and orientation characteristic of aerial scenes. Moreover, 2 images involved debatable object presence, as presented in Table 4. The full breakdown, including filenames, is listed in Supplementary Table 4. Despite its controlled design, DOTA exhibits diverse error modes that could meaningfully distort performance measurements for detectors targeting small or overlapping objects.

3.7.5 Errors identified in Pascal VOC

From a review of 8, 000 images, 46 images in Pascal VOC were flagged with different annotation errors. Localization errors and inconsistent annotations were the most frequent, each affecting 12 images. These typically involved misaligned bounding boxes or varied treatment of edge-case objects like reflections or occlusions. In addition, we observed 10 incorrect labels or non-existent objects, 5 duplicate annotations, and 7 ambiguous or debatable cases, as shown in Table 4. Moreover, all the errors and their corresponding IDs are provided in the Supplementary Table 5.

3.7.6 Errors identified in OpenImage

Open-Image is one of the largest OD datasets. We analyzed 10, 400 samples and identified 240 samples with different types of annotation errors. In this dataset, the most frequent error was multiple annotation errors, which arose in 91 samples. Similarly, the other type of errors, such as incorrect labeling or false positives (87), group annotation inconsistencies (33), inconsistent annotations (8), duplicate boxes (4), localization inaccuracies (14), and ambiguous or debatable examples (3), have also arisen. The summary of all the errors that occur in this dataset is presented in Table 4. Open Images clearly demonstrates that even well-structured, large-scale data collection still has high amounts of annotation errors or noise. All the detected errors with corresponding IDs are provided in Supplementary Table 6.

3.8 Implications on model performance

Annotation errors including missing objects, incorrect labels, inaccurate localization, or inconsistency, might undermine the fundamental reliability of OD systems. These errors create label noise that degrades the quality of ground truth supervision and ultimately adversarial affects performance even for state-of-the-art models by distorting the training signs they depend on to a certain extent. These errors cause inconsistent gradient updates and provide unreliable supervision in the training phase. As the model does not recognize all object instances correctly, missed versus false annotations can lead the model to incorrectly penalize correct predictions or fail to learn from valid instances of objects. This is particularly detrimental for small, occluded, or partially visible objects. Furthermore, group annotation errors and spatial inconsistency lead the model to misinterpret the object boundary or to encode such shortcuts that apply only dataset-specifically, which reduces the model's generalization.

In terms of the model evaluation perspective, annotation noise adversely affects the model's core performance attributes, including the mean average precision (mAP). Missed ground truth objects reduce precision by misclassifying true positives as false positives, and the effect of mislabeled and bounded-box inaccuracy will decrease recall, especially at high Intersection over Union $(IoU) > 0.5$). These problems are intensified with challenging scenarios such as long-tailed distributions (LVIS) or low-data regimes (FSOD), and a small amount of mislabeling can lead to serious metric variance.

In addition to training and evaluation, inconsistencies in annotation may incur damaging, dataset-specific artifacts. Systematic behaviors like the frequent use of "crowd" labels, inconsistent treatments of occlusion, or non-uniform definitions of categories can bias the model's behavior. Detectors learned on such data cannot be robust enough to out-of-distribution and thus may suffer from significant risk when used in safety-critical or real-world scenarios.

3.9 Performance comparison

We evaluate the performance of different state-of-the-art methods over various object detection datasets, as shown in Table 5. These datasets cover a diverse range of real-world categories and scenes from different perspectives. The performance is evaluated on MS-COCO, Pascal VOC, OpenImage, Object365, FSOD, and DOTA datasets. For instance, the performance of the early works on the MS-COCO dataset, like those from Xu et al. (2021)

Table 5 Quantitative performance comparison of representative DVLPE methods on common OD benchmarks

Dataset	Method	Venue,year	mAP	AP
MS-COCO	Xu et al. (2021)	IEEE TIP, 2021	–	21.7
	Chen et al. (2022)	CVPR 2022	–	36.22
	Wu et al. (2023)	ICCV, 2023	–	**53.9**
	Shehzadi et al. (2024)	CVPR, 2024	44.2	47.6
	Wang et al. (2025)	Sensors 2025	43.3	–
PASCAL VOC	Xu et al. (2021)	IEEE TIP, 2021	44.5	–
	Cai et al. (2022)	Electronics, 2022	–	53.5
	Suri et al. (2023)	ICCV, 2023	–	**82.15**
	Shehzadi et al. (2024)	CVPR, 2024	–	65.51
	Wang et al. (2025)	Sensors 2025	51.31	–
Open Images	Veit et al. (2017)	CVPR, 2017	**62.38**	**87.68**
	Peng et al. (2020)	CVPR, 2020	60.9	–
	Liu et al. (2022)	ECCV, 2022	43.14	–
	Ma et al. (2022)	CVPR, 2022	42.9	62.9
	Nou et al. (2024)	MVA, 2024	39.9	71
DOTA	Ming et al. (2021)	Remote Sening, 2021	76.36	–
	Yang et al. (2022)	arXiv, 2022	–	74.4
	Yu et al. (2023)	NeurIPS, 2023	–	**80.61**
	Luo et al. (2024)	CVPR, 2024	49.01	–
	Lu et al. (2024)	NeurIPS, 2024	69.36	–
Object365	Shao et al. (2019)	ICCV, 2019	18.7	**27.3**
	Qi et al. (2025)	IEEE ICASSP, 2025	–	26.6
FSOD	Zhang et al. (2020)	ACCV, 2020	–	29.8
	Fan et al. (2020)	CVPR, 2020	–	**44.1**
	Xiong et al. (2021)	BMVC, 2021	–	29.9
	LaBonte et al. (2022)	WACV 2023	30.6	43

and Chen et al. (2022), had achieved an AP score of 21 to 36. Afterwards, works such as those from Shehzadi et al. (2024) and Wu et al. (2023) exceeded an mAP of up to 40. Most recently, Wu et al. (2023) developed a detector with outstanding performance with an AP of 53.9, which shows the importance of better feature aggregation and task optimizations on large-scale detection. Furthermore, on the PASCAL VOC dataset, previous methods like Xu et al. (2021) achieved AP scores in the mid-40 s, and Cai et al. (2022) reported mAP of 53.5, where the new methods have improved these results tremendously. Shehzadi et al. (2024) reported an AP of 65.51, and have an impressive AP of 82.15 by Suri et al. (2023). This reported AP Suri et al. (2023) and related publications have set a new horizon in this dataset and demonstrate the advantages of better contextual modeling in medium benchmarks. The open Images results show a similar evolution under a much larger label space. Liu et al. (2022), Ma et al. (2022), and Nou et al. (2024) achieve mAP below 44 with AP in the 60–70 range, followed by Peng et al. (2020) with mAP of 60.9, reflecting the difficulty of long-tail categories and noisy annotations. Veit et al. (2017) substantially improve performance to 62.38 mAP and 87.68 AP, indicating that carefully designed training pipelines and localization modules are essential for web-scale detection.

For aerial imagery on DOTA, detectors must handle arbitrary orientations and extreme aspect ratios. Ming et al. (2021) achieves the best overall accuracy with 76.36 mAP while Lu et al. (2024) takes the second best with 69.36 mAP and with an mAP of 49.01 Luo et al. (2024) become the last in comparison on mAP, while Yu et al. (2023) obtains the highest AP at 80.61, suggesting reliable positive predictions followed by Yang et al. (2022) with and AP of 74.4. These results highlight the significance of rotation-aware representations and stable box parameterizations in overhead views. In addition, even in Object365, a large-scale benchmark dataset with detailed annotations, the gaps between different generations of methods are still very large. Shao et al. (2019) with 18.7 mAP and 27.3 AP, and Qi et al. (2025) improved these scores to 26.6 mAP and 26.6 AP, respectively. However, the total numbers remain below those of other datasets, suggesting that Object365 remains a challenging benchmark for scalable and high-recall detection. The FSOD results also reveal the advances in few-shot detection. Early works by Zhang et al. (2020), Xiong et al. (2021) attain mAP $29-30$ with LaBonte et al. (2022) with a slight improvement on the trade-off between mAP 30.6 and AP 43. Fan et al. (2020) achieve the greatest improvement up to 44.1 AP, demonstrating that better meta-learning or feature-transfer methods can significantly improve generalization towards novel categories with less supervision.

Three main dimensions are evident in the different data sets. First, detection performance improves consistently as models advance from basic CNNs to architectures that use more effective pyramids (Ranjbarzadeh et al. 2023; Turab et al. 2022), also deeper context modeling and loss function optimization. Furthermore, popular large-scale benchmarks such as Open Images, DOTA, and Object365 have also played a key role in the development by improving robustness against cluttered scenes, viewpoint variation, and imbalanced distribution of data. Lastly, the observed results on FSOD further highlight an increasing emphasis on scenarios with limited data availability. This trend indicates a general shift towards integrated detection systems, which are still effective in the absence of full annotations.

4 Object detection datasets

To enable the community to conduct more research in OD, the datasets must be publicly available. Datasets play a vital role in the performance of machines and deep learning approaches. A diverse and well-annotated dataset ensures better learning and yields a higher detection rate. Multiple datasets are available in the literature. The most popular datasets, MS-COCO, Open Images, DOTA, Pascal VOC, Object365, KITTI, ImageNet, and FSOD, are covered in this section. A detailed description of these datasets is provided in Table 6, and the year-wise trends of the datasets are shown in Fig. 10.

4.1 Pascal VOC dataset

Pascal visual object classes (P-VOC) (Everingham et al. 2013) is one of the most popular benchmark datasets, featuring a diverse range of scenes from both indoor and outdoor environments. This dataset was developed during the early years of computer vision from 2005 to 2012. The most widely used versions of Pascal VOC are from 2007 and 2012. The earlier version contains 9000 images across 20 different categories, such as person, car, dog, etc., while the updated version, Pascal VOC 2007, includes 11,000 images and 20 distinct object categories with varying poses, scales, and orientations. Each image is annotated with object class labels, bounding boxes, and segmentation masks. Moreover, this dataset is one of the important benchmarks in the computer vision area. The sample images of this dataset are illustrated in Fig. 9.

4.2 MS-COCO dataset

MS-COCO originates from the Microsoft COCO dataset created in 2014, with an updated version proposed in 2017 (Lin et al. 2014). It is one of the most significant and challenging large-scale datasets in the field of computer vision, widely used for object detection and object segmentation tasks. The dataset comprises 330,000 images, which are divided into training, validation, and testing sets. The training and validation sets used for the OD task consist of $123k$ images, and the testing set has $\sim 41,000$ images(without public labels); each image is annotated with bounding boxes, labels, and key information. These samples are categorized into 80 different objects. Compared to Pascal-VOC, the MS-COCO dataset is more challenging, as it includes small objects, dense groups of objects, and occluded objects. The sample images are shown in Fig. 9. Due to the diverse range of objects and the larger number of images, the MS-COCO dataset has become more effective for testing and validating algorithms in real-world scenarios.

4.3 Open Image dataset

Open Image is one of the largest and most recent multipurpose datasets, which includes bounding boxes for 600 object categories with over 9 million labeled images (Kuznetsova et al. 2020). Moreover, segmentation masks were included in the 2019 releases of the dataset. Some other features are also included, such as visual relationship annotations that identify pairs of objects in particular relationships for a more comprehensive understanding. The classes of images include various categories such as coarse-grained and fine-grained

Table 6 Benchmark datasets available for object detection tasks

#	Dataset Name	Venue	Images	Instances	Description
1	MS-COCO (Lin et al. 2014)	ECCV 2014	∼123K	∼850k	Large-scale dataset for object detection and scene understanding in context
2	DOTA (Ding et al. 2021)	TPAMI 2021	∼11K	∼1.7M	Large-scale aerial dataset with oriented bounding boxes for object detection
3	Open Images (Kuznetsova et al. 2020)	IJCV 2020	1.9M	∼9 M	Comprehensive dataset for diverse computer vision tasks with rich annotations
4	Pascal VOC (Everingham et al. 2013)	ECCV 2012	∼11K	∼27K	Benchmark dataset for object detection and segmentation in natural scenes
5	Object365 (Shao et al. 2019)	1CCV 2019	638k	10.1M	Object365 aims to encourage research in object detection, particularly for various objects found in the wild
6	FSOD (Kang et al. 2019)	CVF 2019	100K	300K	The FSOD dataset is a benchmark for testing few-shot object detection on novel classes with limited training samples
7	ImageNet (ILSVRC) (Russakovsky et al. 2015)	IJCV 2015	∼1.2M	∼1.5M	A large-scale image dataset structured by the WordNet hierarchy, used to benchmark object recognition, detection, and localization models
8	KITTI (Geiger et al. 2012)	CVPR 2012	∼22K	∼80K	A real-world autonomous driving dataset collected with stereo cameras, LiDAR, and GPS/IMU, featuring benchmarks for 2D and 3D vision tasks
9	Mini6KClean (Tong and Wu 2023)	JVCIR 2023	6K	∼55K	It is the clean updated version of MS-COCO dataset. In this dataset, labeling errors of the MS-COCO were corrected
10	Sama-COCO (Zimmermann et al. 2023)	arXiv 2023	∼123K	∼1.1M	It is a re-annotated version of the MS-COCO dataset with higher quality labels and compares it against the original
11	COCO-OI (Borji xxx)	ICLR 2023	∼398K	∼1.5M	A re-annotated version of MS-COCO and OpenImages that identifies and corrects annotation errors, producing a refined dataset
12	MJ-COCO (Kim et al. 2025)	arXiv 2025	∼123K	∼1.2M	It is an enhanced version of MS-COCO-2017, with annotation errors automatically corrected using model-driven methods
13	iSAID (Waqas Zamir et al. 2019)	CVPR Workshop 2019	∼2.8K	∼655k	A large-scale aerial dataset based on DOTA images, introduces richly annotated, high-resolution image patches to improve fine-grained localization and boundary precision in remote sensing OD.
14	AI-TOD (Wang et al. 2021)	ICPR 2021	∼28K	∼282k	A specialized benchmark targeting tiny object detection in aerial scenes. Consisting of extremely small instances less than (16±16 pixels) across 8 categories.
15	DIOR (Li et al. 2020)	ISPRS Journal of Photogrammetry and Remote Sensing 2020	∼23.4K	∼190k	A diverse optical remote sensing dataset with 20 object categories captured from multi-resolution satellite sensors. Includes notable variations in geography, season, and viewpoint.

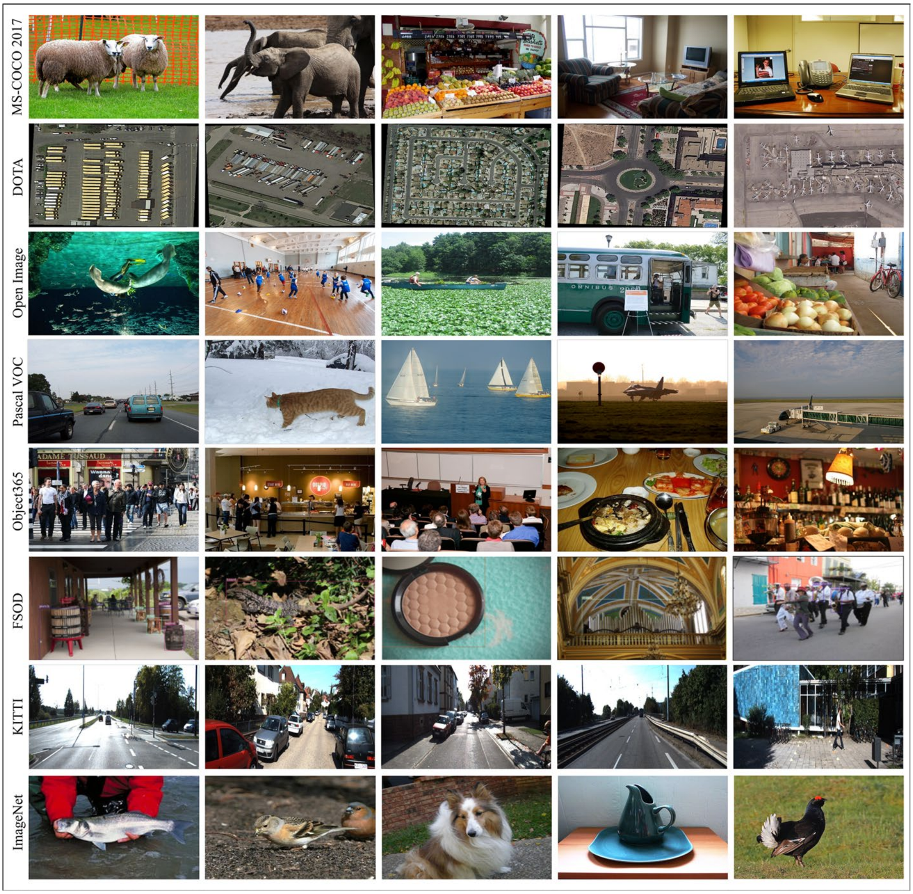

Fig. 9 An illustration of sample images from each dataset, highlighting their diversity

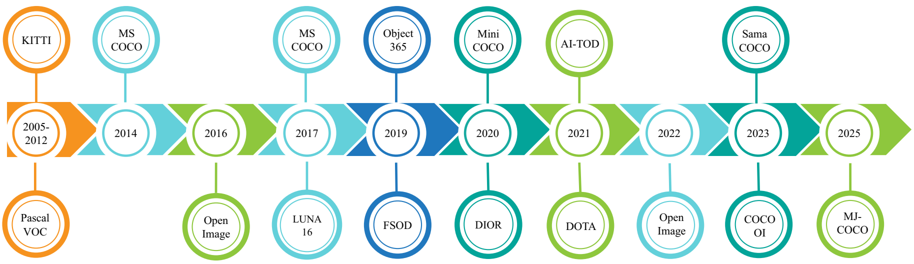

Fig. 10 Evolution of popular object detection and image annotation datasets from 2005 to 2025. The timeline illustrates key milestones, including the release of Pascal VOC, MS COCO, Open Images, FSOD, and emerging datasets such as Sama-COCO and MJ-COCO

object classes, scenes, events, as well as materials and attributes like texture and color. The names given to the classes of images are segregated into positive and negative labels. Positive labels are used for cases in which instances are present in the image, while negative labels indicate the absence of instances. Research has demonstrated that when classifiers are applied to the dataset, both with and without negative labels, the inclusion of negative labels significantly improves classification accuracy.

4.4 ImageNet (ILSVRC) dataset

The ImageNet large scale visual recognition challenge (ILSVRC) is a large, challenging dataset that has been utilized in the competition from 2014 to 2017 to advance CV technology (Russakovsky et al. 2015). ILSVRC-2014 and ILSVRC-2017 are the most commonly used versions of the dataset. These two datasets share the same training and validation sets but have different test sets: the training set consists of approximately 456,000 images, the validation set contains about 20,000 images, and the testing sets have 40,000 images for ILSVRC-2014 and 65,000 images for ILSVRC-2017. This dataset is used for both classification and detection tasks. For the classification task, there are more than 20,000 image categories, with a large number of images in each category, while for the detection task, nearly 200 object categories are used. Each image is assigned a primary category label along with several potential secondary labels.

4.5 FSOD dataset

The few-shot object detection (FSOD) dataset is collected from existing large-scale OD datasets such as ImageNet and OpenImage (Kang et al. 2019). It streamlines the labeling systems of these datasets by merging the leaf labels found in their original label trees. Labels with similar meanings, such as "ice bear" and "polar bear," are categorized together, while any semantics that do not align with the leaf categories are eliminated. The dataset contains a total of 1000 classes, with 496 taken from the OpenImage and 531 from the ImageNet datasets. The dataset is divided into training and testing sets, where the training set includes 52, 350 images, 147, 489 bounding boxes, and 800 classes, while the testing set has 200 classes, 14, 152 images, and 35, 102 bounding boxes.

4.6 DOTA dataset

The dataset of O in Aerial images (DOTA) is the largest dataset for the OD task involving aerial images (Ding et al. 2021). This dataset includes three versions: DOTA-V1.0, DOTA-V1.5, and DOTA-V2.0. DOTA-V1.0 contains 2806 large aerial images of various sizes, 15 different classes (i.e., Plane, Baseball diamond, Basketball court, etc.), and 188, 282 bounding boxes. In 2019, a new version DOTA-V1.5 was released with one additional new class (container crane). This version contains 2, 806 images with 402, 089 bounding boxes and 16 classes. Furthermore, the latest version DOTA-V2.0 was released in 2021. The data were collected from various sources, such as JL-1, GF-2 satellite images, and Google Earth images. This version includes two new categories (airport and helipad), bringing the total to 18 categories, 11, 268 images, and 1, 793, 658 bounding boxes. Compared to other versions, it was more stable during training and more challenging overall.

4.7 KITTI dataset

The KITTI dataset was first released by the Toyota Institute of Technology and the Karlsruhe Institute of Technology in 2012 (Geiger et al. 2012). The KITTI dataset is meant to collect data using an acquisition vehicle that captures data in urban, highway, and rural areas. Each frame contains 15 cars and 30 people of different occlusion and truncation levels. The dataset includes 389 pairs of stereo images, optical flow maps, 39.2 km of visual range sequences, 15,000 point cloud frames, and an impressive 200,000 manually labeled 3*D* object frames. The data acquisition vehicle is fitted with two grayscale cameras, two color cameras, a Velodyne 64-line LiDAR, four optical lenses, and a comprehensive GPS navigation system. KITTI offers raw data for 3*D* perception benchmarks, evaluation metrics for each benchmark, and an online platform to test and compare various methods.

4.8 Object365 dataset

Object365 is one of the largest datasets featuring 365 different categories, primarily collected from Flickr (Shao et al. 2019). First, the most common and diverse eleven super categories are selected. These categories include electronics, kitchen, clothes, food (vegetables), transportation, office supplies, bathroom, living room, animals, humans, related accessories, and instruments. The super categories are then further divided into 442 existing categories related to daily life. Some of these categories are rarely represented, but in the initial phase, all of them are annotated in 100*k* images, and subsequently, the most frequent 365 object categories are selected. The final version consists of 638*k* images and 10.1*M* annotations.

4.9 Mini6KClean dataset

Mini6KClean is the refined version of the Mini6K dataset, which is a subset of the MS-COCO 2017 dataset (Tong and Wu 2023). Mini6K was created from the train2017 set of the MS-COCO dataset, comprising 6000 images. It contained annotation errors, such as missing or incorrect labels, in over 12% of the images. To resolve this issue, a new version, Mini2022, has been introduced. It is a cleaned subset of 729 images with corrected annotations, which was integrated back into Mini6K to create Mini6KClean. Mini6KClean includes 6000 images and 55, 111 object instances. Mini6KClean has 10, 894 more instances than the original Mini6K.

4.10 Sama-COCO dataset

Sama-COCO is the re-annotated version of the MS-COCO 2017 dataset (Zimmermann et al. 2023). It was developed by Sama's in-house team to improve the quality of MS-COCO 2017 annotations for object detection and segmentation tasks. Sama-COCO introduces several improvements to the original MS-COCO dataset while maintaining the same set of 123, 287 images and 80 object categories. Sama-COCO enhances annotation precision, refines crowded annotations by decomposing them into individual instances, and increases the number of object instances to approximately 1.12 million, which is 218, 000 more than the original MS-COCO 2017 dataset.

4.11 COCO-OI dataset

COCO-OI is constructed from a combination of the MS-COCO and OpenImage datasets (Borji xxx). It focuses on the 80 instance categories that are common to both datasets. While maintaining COCO-style labeling quality, COCO-OI introduces a domain shift by integrating images from OpenImage that are not present in MS-COCO. This allows objects to appear in a wider range of scenarios, positions, lighting conditions, and backgrounds, exposing models to greater variation. It contains approximately $398k$ images and $1.2M$ instances. It exclusively employs bounding boxes and does not utilize segmentation masks.

4.12 MJ-COCO dataset

MJ-COCO is the latest re-annotated version of the MS-COCO 2017 dataset. This dataset addresses common annotation issues found in the original MS-COCO datasets, such as missing object labels, inaccurate bounding boxes, and duplication, while maintaining the same set of 123,287 images and 80 object categories. The MJ-COCO dataset has been collected using a four-stage pseudo-labeling refinement pipeline to systematically correct these problems without requiring manual annotation. They identified 52, 621 anomalous images in the MS-COCO dataset and correctly reannotated them in the MJ-COCO dataset.

4.13 Medical imaging object detection datasets

Although segmentation dominates in medical imaging, several benchmark datasets provide bounding-box-based annotations suitable for object detection tasks. These datasets enable detection models, such as Faster R-CNN, YOLO, and DETR, to localize lesions, lung nodules, or abnormalities without requiring dense pixel masks. DeepLesion (Yan et al. 2017) is one of the earliest large-scale medical detection datasets, containing over 32000 CT slices annotated with lesion bounding boxes. However, subsequent studies, including Lesion Harvester (Cai et al. 2020) and Bai and Xia (2022), revealed that DeepLesion suffers from missing annotations, with more than 50 % of lesions unlabeled, and inconsistent lesion tags across series. More recently, in 2025, Erickson et al. (2022) further reported class imbalance and missing body-part tags, confirming annotation incompleteness as a persistent issue. LUNA16 (Setio et al. 2017) focuses on lung-nodule detection in CT images, providing manually validated bounding boxes. While widely used, it also exhibits variability in annotator agreement and borderline nodules near the detection threshold. VinDr-Mammo (Nguyen et al. 2023) extends detection to chest X-rays and mammograms. Each image includes bounding boxes around pathological findings. Despite expert curation, later evaluations identified labeling inconsistency and scale bias for small findings. These domain-specialized datasets highlight that even in high-stakes domains, incomplete, noisy, and inconsistent annotations remain key obstacles for reliable detection.

4.14 iSAID dataset

The iSAID (instance segmentation in aerial images dataset) is a large-scale benchmark specifically designed for instance segmentation and object detection in high-resolution aerial imagery (Waqas Zamir et al. 2019). It provides a more realistic depiction of object boundaries by introducing pixel-accurate instance masks, although it is based on the same underlying imagery as the DOTA-v1.0 dataset. iSAID contains 2,806 high-resolution aerial images with over 655,451 densely annotated instances across 15 object categories, including planes, ships, storage tanks, roundabouts, bridges, and small-scale vehicles. All images in the iSAID dataset are in a patched format, specifically 800×800 crops, to ease the load on the system during training and evaluation on high-resolution images, which are $4,000 \times 4,000$ pixels large. The dataset is perfect for assessing and annotating the inconsistencies, such as boundary uncertainty, fine-grained localization, and class confusion issues that are also discussed in the annotation errors of the survey, as iSAID's segmentation masks reduce the ambiguity of the objects' extent and shape compared to DOTA's oriented bounding boxes. Moreover, iSAID is an additional, more detailed granularity benchmark to DOTA, which is crucial in the evaluation of remote-sensing detectors that are especially concerned with the detection of irregularly shaped objects, objects that are occluded, and objects that are placed in highly dense scenes.

4.15 AI-TOD

The tiny object detection in aerial images (AI-TOD) is a specialized benchmark dataset addressing a major challenge in remote-sensing OD systems by targeting tiny objects in aerial and satellite images (Wang et al. 2021). Tiny objects, often only a few pixels in size, are highly sensitive to annotation noise, localization errors, and inconsistent labeling, which aligns directly with the error categories analyzed in our survey. AI-TOD contains over 282, 000 annotated instances across eight categories, such as vehicles, ships, small aircraft, and containers, captured from diverse aerial platforms under varying illumination, density, and clutter conditions. This dataset is one of the most difficult standards for assessing detectors that must function under large-scale fluctuations because over 70% of items fall below the size threshold of 16×16 pixels. The dataset includes both training and testing splits with standardized annotation formats using horizontal bounding boxes. Many recent works studying label noise in remote sensing adopt AI-TOD due to its sensitivity to even minor localization discrepancies. Its high density, clutter, and object-scale imbalance make it an ideal dataset for validating noise-robust detection algorithms, semi-supervised pipelines, and data-centric annotation refinement methods.

4.16 DIOR dataset

The DIOR (dataset for object detection in optical remote sensing images) (Li et al. 2020) dataset is a large-scale benchmark consisting of 23,463 images with 190,000 labeled object instances across 20 diverse categories, including airplanes, vehicles, ships, storage tanks, playgrounds, harbors, and airports. Images are sourced from multiple satellites with varying resolutions $(0.5 - 30m)$, leading to significant diversity in viewpoint, scale, and appearance. DIOR incorporates seasonal changes, geographical variations, and different weather conditions so that they can try to simulate all of the complexities encountered in actual remote sensing. DIOR is not focused on DOTA since it applies axis-aligned bounding boxes. That is a different type of annotation style that illustrates the impact of box alignment on localization errors and evaluation metrics, which we have extensively discussed in our surveys. An extension called DIOR-R also tries to improve constant rotational invariance for detection metrics by adding oriented bounding boxes. Thus, DIOR is a great dataset to tackle horizontal and rotated detection problems. Its scale, diversity, and multi-sensor coverage make it one of the most influential datasets in the remote-sensing OD community.

5 Challenges in datasets and future directions

This section provides detailed descriptions of the annotation errors and challenges in the OD datasets and proposes future solutions for overcoming these challenges.

5.1 Errors in object detection datasets

The performance of object detection systems depends on the quality of the model's predictions and the quality of the annotations in the dataset. Benchmark datasets contain systematic errors, and this can affect the performance of OD systems. Such errors will not only affect a model's training and evaluation but also affect the understanding of the model's capabilities in the real world. In this section, the errors found in MS COCO, OpenImage, PASCAL VOC, DOTA, LVIS, Object365, and FSOD are discussed in detail, and also provide the possible consequences on performance. In Table 7, a summary of the errors identified in the seven most important OD datasets is presented, with (✓) indicating the presence of the error type in the dataset and the (✗) indicating that the error is not present.

5.1.1 Missed annotations

Missed annotations are one of the most common annotation errors in OD datasets. It refers to instances when the objects in an image are present but are completely unlabeled in the ground truth. Such omissions are particularly frequent when there are small, covered, trimmed, or small objects that are easily missed during the annotation process. Although these objects are visible, they are overlooked during the annotation process, which leads to a decrease in the model's performance. These oversights are considered false positives during evaluation and result in underestimating a model's actual performance. This is the most common type of error in the COCO validation set and significantly affects the AP, in particular for smaller object sizes.

Table 7 Comparison of common annotation error types across popular object detection datasets

#	Error Types	MS-COCO	DOTA	LVIS	Open Images	P-VOC	Object365	FSOD
1	Missed Annotaions	✓	✓	✗	✗	✓	✓	✓
2	Incorrect Labels	✓	✓	✓	✓	✓	✓	✓
3	Localization Error	✓	✓	✓	✓	✓	✓	✓
4	Duplicate Annotations	✓	✓	✗	✓	✓	✓	✓
5	Inconsistent Annotations	✓	✓	✓	✓	✓	✓	✓
6	Debatable Examples	✓	✓	✓	✓	✓	✗	✗
7	Group Error	✓	✗	✗	✓	✗	✗	✗

5.1.2 Incorrect labels and non-existing objects

This category includes two types of errors: class mislabeling and false positives. Class mislabeling happens when an existing object is assigned the wrong category class. For example, a goat is categorized as a cow. False positives happen when sections of the background or some unstructured patterns get mistakenly annotated as objects. Such errors lower the quality of the training data and lead the model in the wrong direction during learning and evaluation. Such errors compromise inter-class discrimination, making it difficult to train detectors that accurately reflect object semantics. As shown in Fig. 11a, where ambiguous objects are either annotated inappropriately or labeled as inaccurate.

5.1.3 Localization errors

Localization errors are those types of errors where the annotated bounding boxes do not align with the objects correctly. These inaccuracies can result in bounding boxes that are either too big, extending outside the object's boundary, or are too small, underrepresenting the object, and cutting portions of the object. These issues would affect IoU evaluation metrics and can lead to false negative or false positive detections at higher IoU precision thresholds. Some examples of localization errors are in Fig. 11b, where bounding boxes cut off excessive amounts of portion from the object. These mismatches degrade spatial accuracy and confuse the model training signals. These problems reduce the boundary precision and clarity of the detections across different detection frameworks. These mismatches reduce the boundary precision and clarity of the detections across different detection frameworks.

5.1.4 Duplicate annotations

Duplicate annotation errors occur when multiple bounding boxes are drawn around the same object or when a single large bounding box encompasses a group of distinct objects. These duplicates can artificially inflate object counts and mislead the evaluation protocol, resulting in the underreporting of a model's precision. Examples of overlapping or excessively bounding boxes are shown in Fig. 11c. If not properly handled during training or evaluation, these annotations can cause models to learn ambiguous object representations and lead to penalization of otherwise valid predictions.

5.1.5 Inconsistent annotations

Inconsistent annotations refer to variations in labeling policies across different images for the same object types or visual scenarios. For example, reflections, statues, or images of people in posters may be annotated in some instances but ignored in others. This inconsistency introduces noise into the training data and evaluation set, diminishing reproducibility and model robustness. As illustrated in Fig. 11d, such inconsistencies are evident in cases involving mirrored reflections or ambiguous visual representations. For example, a person and their reflection in a mirror are labeled in some images, while in another sample, a cat is not labeled in the mirror. Additionally, in some cases, a person on the poster is labeled, while in others, the person is not labeled, as shown in Fig. 11d.

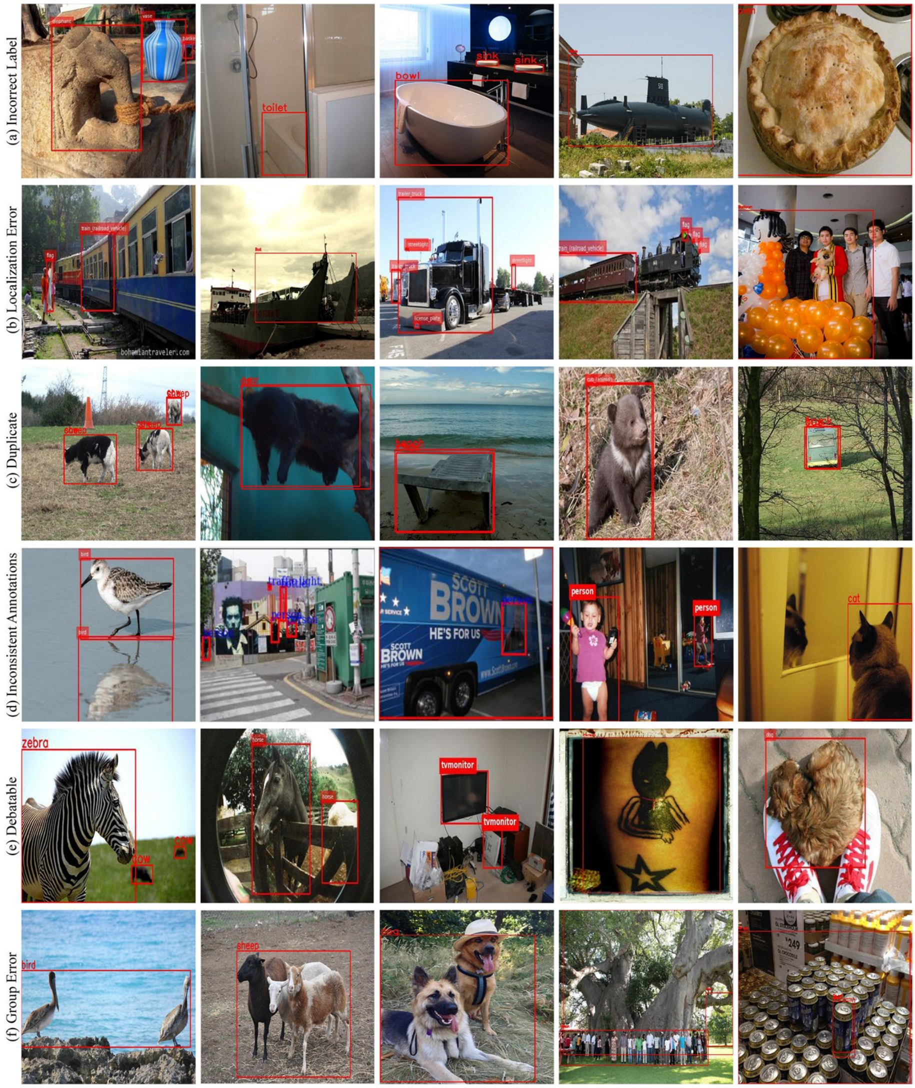

Fig. 11 A collection of images illustrating challenges in OD dataset annotations, where object labels, positions, and categories often vary or conflict across examples

5.1.6 Challenging or debatable

These are one of the most challenging types of errors, where certain categories are hard to annotate even for humans. For instance, images may have low resolution, show occlusions, depict ambiguous scenes, or contain other elements that make annotation difficult. In these cases, human annotators might disagree on whether an object is present or what label to assign. These samples introduce uncertainty into the training and evaluation processes and tend to cause high variability in agreement levels among different annotator teams. Figure 11e displays examples of such difficult annotation cases, including partially visible and ambiguous animals. These images demonstrate the limits of human perception, showing that not all mistakes are the result of negligence or inconsistency, and that visual ambiguity plays a significant role.

5.1.7 Group errors

Group errors are those errors in which the same or different multiple objects are located near each other and are incorrectly annotated with a single bounding box. It is essential to recognize that a common mistake in this context is to encircle a group of objects with one bounding box instead of assigning individual bounding boxes to each object, as shown in Fig. 11f. Bounding boxes should be given to each individual object to enhance annotation quality.

5.1.8 Dataset diversity and its impact on annotation reliability

The effectiveness of models in object detection and the reliability of annotations are significantly influenced by the diversity of datasets. Labeling errors arise from the class distribution, light conditions, environmental background, and other contextual variables. Dominant classes such as person and car cause a class imbalance, while other, rarer classes suffer from under-annotation, creating more false negatives and restricting the ability of the model to make accurate predictions. Another factor, such as environmental changes including lighting and weather variation, can also affect the annotation consistency and model performance. According to Kumar et al. (2024), Chandio et al. (2021), and Kumar et al. (2021), maintaining diverse, balanced, and augmented datasets improves annotation reliability and generalization across domains. Highlighted that keeping a variety of well-augmented datasets enhances generalization across domains, while Singh et al. (2023) demonstrated that context-rich and adaptive datasets improve system reliability under changing circumstances. To achieve varied real-world scenarios with consistent labeling, we need diverse performance, well-balanced category representation, and environment-aware annotation and augmentation. The interpretability of performance measurements, benchmarking fairness, and model training fidelity are all affected by these error types. As datasets continue to grow in size and diversity, the requirement for standard annotation auditing and modification becomes more essential.

5.2 Recommendations and future research directions

To enhance the robustness and reliability of object detection systems, future work should also consider efforts to prevent the occurrence of annotation errors during the data annotation or dataset creation phase, rather than only trying to identify errors after training the model. This is particularly relevant for the growing field of data-centric AI, as data quality has become the main focus for model developers, compared to just improving model architecture, as discussed by research pioneers like Andrew Ng in 2021.[1] Unlike the traditional model-centric approaches, which focus broadly on the algorithms themselves, data-centric AI emphasizes the need for representative, consistent, and well-annotated datasets in AI models. In OD, this involves rethinking how datasets are built, how models interact with

[1] https://landing.ai/data-centric-ai.

noisy labels, and how to improve annotation tools to reduce inconsistencies. The directions below highlight key areas for future innovation in this domain.

5.2.1 Development of benchmark datasets with verified annotations

It is important to create benchmark datasets to have verified and traceable labels. Existing datasets include annotation errors, including inconsistent labels, localization errors, and missing annotations, which mislead the AI model training and evaluation processes. Datasets in the future must be created using more precise methodologies and multi-stage annotation processes. This process includes multiple annotators, going through multiple review cycles, and clear annotation correction protocols. Moreover, using a human in the loop process for the annotations and error correction can help to maintain the high-quality annotations. Also, keeping records of the annotated datasets that include revisions, logs of annotated disagreements, and differing opinions may improve the quality of the benchmark test.

5.2.2 Standardized annotation quality metrics and evaluation protocols

Another field of interest is developing standardized evaluation protocols and metrics for assessing annotation quality. The quality of annotation is undeniably important, but the field still lacks a common metric for measuring how clean and reliable a dataset is. Future work should aim toward the development of quality annotation assessment systems that measure and identify the level and severity of annotation issues, such as missing labels, spatial localization, and ambiguous instances. Moreover, similar to mAP for model evaluation, assessment systems for datasets that measure quality and reliability, labeled ‘Annotation mAP’ or ‘label precision score’, would enable and improve the ability of the academic and practitioner community to make effective and efficient comparisons of datasets and target prioritization for dataset refinement and cleaning.

5.2.3 Context-aware and quality-assured annotation tools

Another possible direction is the creation of context-aware, trusted annotation tools. Particularly in complicated scenarios or when dealing with partial object visibility, annotators sometimes operate with little information, naming objects within isolated image patches or frames, which results in errors. Features like zooming, multi-resolution viewing, and contextual image previews across image sequences or datasets should be included in future annotation tools. Additionally, these tools can include instance history, which enables annotators to reference earlier labeling decisions for related instances. Moreover, AI-based recommendations using model predictions can serve as a second layer of protection against potential labeling errors, ensuring label consistency. Importantly, these tools also need to provide a visibility estimate, with what fraction (or percent) of the object is visible in the image, and an auto-flag of borderline cases where annotation decisions may vary. It would offer real-time instructions about determining whether a partially visible object needed to be annotated, using prespecified thresholds. The most important concern is that such tools be extensively tested and validated before public release in order to avoid introducing new annotation errors and propagating the behavior of model bias.

5.2.4 Active annotation refinement via model feedback loops

Enhancing annotation quality involves collaboration between humans and models, which can be integrated into an ensemble framework to produce more reliable annotations. Active annotation with model feedback is a promising direction. During training, models may detect outliers, contradictions, or high uncertainty predictions and suggest that these cases be re-annotated. Methods such as active learning can concentrate on the most informative or suspect samples, whereas those based on ensemble disagreement or self-supervision may be able to detect weak supervised signals. The collaboration allows the annotation pipeline to be more flexible and data efficient by making it possible for both the labels and the performance of models to improve iteratively.

5.2.5 Learning robust models under noisy and incomplete annotations

As collecting such high-quality datasets with perfect annotations is not often viable for large-scale or specialized domains, the future work should also aim to build models that can provide robust learning based on noisy and incomplete annotations. This ranges from learning in a noise-aware manner, like uncertainty modeling and outlier-robust loss functions, to curriculum learning. In addition, models need to be able to learn well with weak supervision and multiple types of labels to facilitate generalization. Semi-supervised and self-supervised pretraining methods have shown noticeable potential in this respect, which could act as a bridge for the gap between scarce annotated data to real-world performance requirements.

5.2.6 Cross-dataset annotation harmonization and transferability

The other thing is cross-dataset harmonization with respect to their transferability. Object detection models are trained on more and different datasets; there may be little variability or potentially a conflict between the labeling criteria (box size acceptance, category definition), which can reduce the generalization. There should be future work to study the impact of these label inconsistencies on performance and how protocols determine the alignment of annotations across datasets. Methods to align ontology, normalize bounding boxes, and style translation would be useful for developing cross-domain, foundation-level object detectors that are universally applicable to diverse labeling conventions.

5.2.7 Standardizing bounding box definitions and label hierarchies

A major challenge of cross-dataset inconsistency is that there are no 'standards' or universally accepted protocols for bounding box definition and class taxonomy. In addition, bounding box tightness, occlusion or truncation representation, and rotated boxes vs. segmentation-based ones deviate across datasets, causing differences in localization. In addition, the categories could also be defined differently from one dataset to another. For example, some datasets present a simple basic list of labels (e.g., COCO) while others use more fine-grained hierarchical taxonomies (e.g., LVIS and Open Images). This kind of variation can make category definitions inconsistent. To minimize annotation noise between datasets, there is a need for standardized bounding box geometry protocols or the usage of a common/compatible label hierarchy. Such measurements would help to align datasets with annotations, especially in the case of training these foundational object detectors.

5.2.8 Ethical and fairness-aware annotation standards

Annotation quality is also an ethical value, especially in people-centric applications, so future work should focus on fairness-aware annotation practices. Errors in annotation often show bias against underrepresented or sensitive groups, such as people of various demographics, abilities, and social statuses. Dataset creators must use clear annotation guidelines that account for biased sources and implement quality control measures to filter demographic or category-related inconsistencies. Fairness auditing of annotation and evaluation processes will be vital to ensure that models trained on these datasets are inclusive, fair, and safe to deploy in the real world.

5.2.9 Foundation-model-assisted and human-in-the-loop error correction pipelines

In the future, it would be interesting to investigate the more seamless integration of dataset error identification into new foundation models and large-scale annotation workflows. Vision-language and segmentation base models (e.g., CLIP-, DeTR- or SAM-like methods) may then act as separate “label auditors” to supervise the semantic consistency between image regions and class names, improving the quality of bounding box extents, or bringing attention to potentially missing or mislocalized objects. Additionally, error identification could be integrated into automatic annotation pipelines as active learning and crowd-sourcing. Instead of uniformly annotating data, models should highlight for human or crowd workers the most suspicious or samples of highest uncertainty through signals based on disagreement or confidence scores, allowing them to re-label these samples. This joint contribution of foundation-model assistance, automatic filtering, and human-driven feedback offers a promising means for achieving continual improvement in object detection dataset quality at scale.

5.2.10 Defining visibility thresholds for object annotation

Standardized guidelines for annotating the visibility of occluded or partially visible objects are often ignored in datasets. Some datasets annotate a small visible part of an object (e.g., hand or foot) as the whole object (Person), while other datasets leave it unannotated due to a lack of visual or texture features. This inconsistency causes confusion during training and evaluation, as models receive ambiguous supervision and ground-truth expectations may differ across datasets. To resolve this issue, future datasets should focus on establishing visibility thresholds to determine whether an object may be annotated or removed. For instance, annotators might be tasked to label objects only if more than a certain fraction (e.g., 30% or 50%) of the object is fully visible according to quantifiable visual measures. Annotation guidelines should also specify how occlusion, truncation, or scale affects a labeling decision consistently. Furthermore, by defining such thresholds, subjectivity would be reduced and improved inter-annotator consistency between datasets. Visibility-aware annotation protocols are particularly important in safety-critical domains like autonomous driving, robotics, and surveillance, where partially visible objects like a pedestrian appearing from behind a parked car or an approaching vehicle at the blind spot should be detected as early as possible to reduce security risk. A systematic and detailed annotation for such cases has a direct effect on the reliability of models deployed in real-life, high-risk scenarios.

Addressing these types of annotation errors in OD systems demands a shift from model-centric methodologies toward data-focused approaches, while focusing on the importance of annotation correctness, uniformity, and clarity. The quality and reliability of datasets are two key factors influencing the performance of modern detection models. Such issues have potential impact beyond benchmark testing to safety-critical domains such as autonomous perception and medical imaging, where even marginal labeling errors could represent large real-world risks. For example, Chandio et al. (2022) emphasizes the need for precise bounding-box calibration to achieve better localization in challenging scenarios, and Khan et al. (2022) proposes a noise-robust pseudo-example generation strategy to enhance model adaptation ability even with noisy labels. Similarly, Kumar et al. (2024) provides a comprehensive analysis of image data augmentation and highlights the importance of diverse and well-refined data for consistent generalization across domains. Future work should also target the availability of valuable, version-controlled datasets, and annotation quality standardization and verification metrics, along with smart annotation systems able to automatically check visibility and context-dependent validation. Incorporating strong learning strategies to handle noisy/incomplete annotations, as well as constant cross-dataset normalization, will be required for fair, reproducible, and stable detection accuracy. How to achieve better object detection models that are more accurate and more interpretable, pushing towards deployable trust?" A systematic annotation audit for enhancing data quality can advance the search for reliable object detection methods fit for real-world applications.

6 Conclusion

In this survey, we first described the applications, structures, and coverage of our review on annotation error for object detection benchmarks. We emphasized the crucial role of instances with accurate annotations, especially in localization quality, for training and testing the OD models correctly. The survey identified the most common annotation errors in benchmark datasets, such as incorrect bounding boxes, missing labels, and category mismatches etc. By investigating existing error detection mechanisms closely, we investigate how these annotation errors are identified and validated in the literature. Moreover, we manually investigated a number of popular datasets to obtain insights into the nature of annotation issues in practical scenarios. These findings also serve to support the argument that data quality is a persistent challenge for object detection, and the correction of these errors is crucial for improving both dataset reliability and model performance.

Moreover, future research will incorporate more sophisticated techniques for finding and correcting annotation errors, especially in large datasets. Advanced methods like self-supervised learning and weak supervision open up new possibilities towards reducing the human annotation costs under large-scale dataset collection. The usage of synthetic data generation is also known to play an important role in reducing the annotation errors by generating various high-quality datasets that complement real-world information. Model-centric and data-centric AI approaches can also work together to improve both the efficiency of the error detection workflow as well as dataset quality across domains. Together with automatic error-correction workflows, these technological advances will be crucial to overcoming the increasing issues raised by annotation noise and dataset irregularities.

Furthermore, this survey not only presents an extensive analysis of annotation errors of OD benchmarks but also motivates future research to develop novel solutions for data quality enhancement, error-aware training, as well as dataset improvement. With further improvements in the error detection and validation techniques, the research community should make even more robust, reliable, and scalable object detection systems.

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Declarations

Conflict of interest The authors declare no Conflict of interest.

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Authors and Affiliations

Adnan Hussain[1] · Kaleem Ullah[1] · Muhammad Afaq[1] · Muhammad Munsif[1] · Altaf Hussain[1] · Sung Wook Baik[1]

✉ Sung Wook Baik
sbaik@sejong.ac.kr

Adnan Hussain
adnanhussain@sju.ac.kr

Kaleem Ullah
kaleemullah@sju.ac.kr

Muhammad Afaq
Muhammadafaq@sju.ac.kr

Muhammad Munsif
munsif@sju.ac.kr

Altaf Hussain
a.hussain@ieee.org

1 Sejong University, Seoul 143-747, Republic of Korea